

# Cross-Wearer Gesture Recognition — LOUO Variant Comparison

Generated 2026-05-12 21:06:52

## Data & configuration

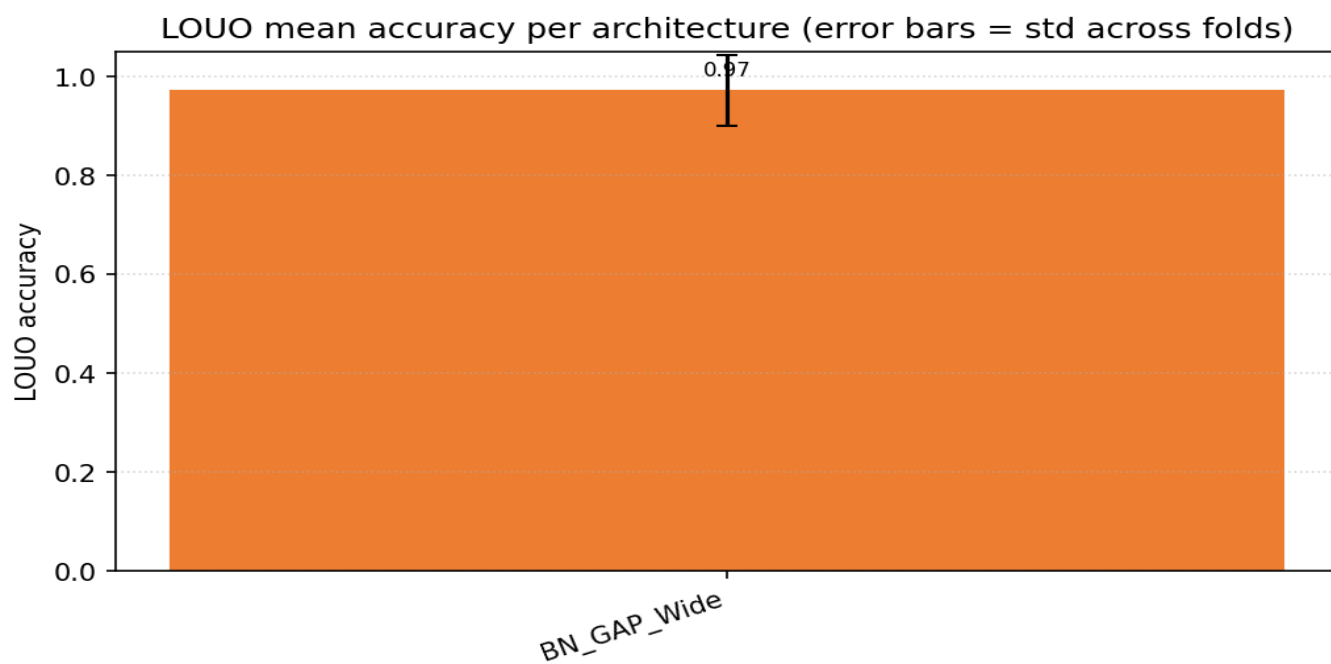
|                             |                                                                                                                        |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------|
| LOUO path 1                 | /home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic                                                                  |
| LOUO path 2                 | /home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic                                                                  |
| LOUO path 3                 | /home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic                                                                |
| LOUO path 4                 | /home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic                                                                |
| LOUO path 5                 | /home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic                                                                 |
| LOUO path 6                 | /home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic                                                                |
| LOUO path 7                 | /home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic                                                                |
| LOUO path 8                 | /home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic                                                               |
| LOUO path 9                 | /home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic                                                            |
| LOUO path 10                | /home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic                                                               |
| LOUO path 11                | /home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic                                                            |
| LOUO users loaded           | 11 (11_Mansh, 12_Marcus, 14_StephenV2, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry, 8_Jestin) |
| Excluded classes            | Double_Nothing                                                                                                         |
| Classes                     | 6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll                            |
| Sensor channels             | 176 (sequence length 90)                                                                                               |
| Resample / filter           | 90 steps · Butterworth cutoff 10.0 Hz, order 4                                                                         |
| Per-trial normalisation     | zscore                                                                                                                 |
| Downstream normalisation    | minmax                                                                                                                 |
| Sampling rate               | 30.0 Hz                                                                                                                |
| Random state                | 42                                                                                                                     |
| Augmentation                | on · copies=2 · amplitude_scale                                                                                        |
| Epochs · batch size         | 40 · 16                                                                                                                |
| Early stopping              | on · monitor val_loss · patience=8                                                                                     |
| LOUO inner-validation       | user                                                                                                                   |
| LOUO variants               | BN_GAP_Wide                                                                                                            |
| Final-retrain variant       | BN_GAP_Wide                                                                                                            |
| Final-retrain held-out user | — (trained on all users)                                                                                               |

## Variant comparison (LOUO)

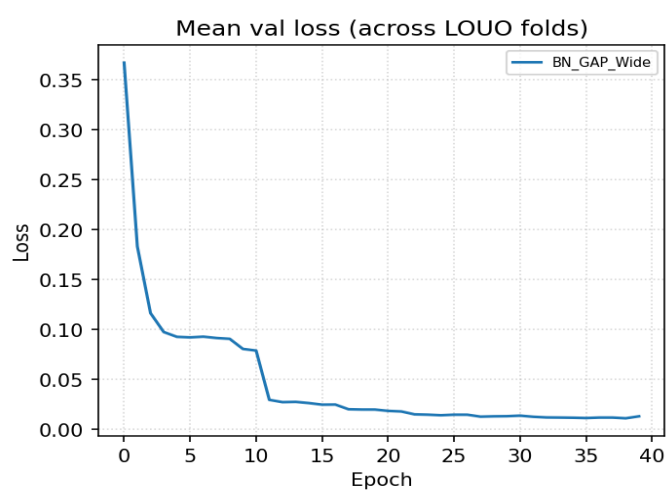
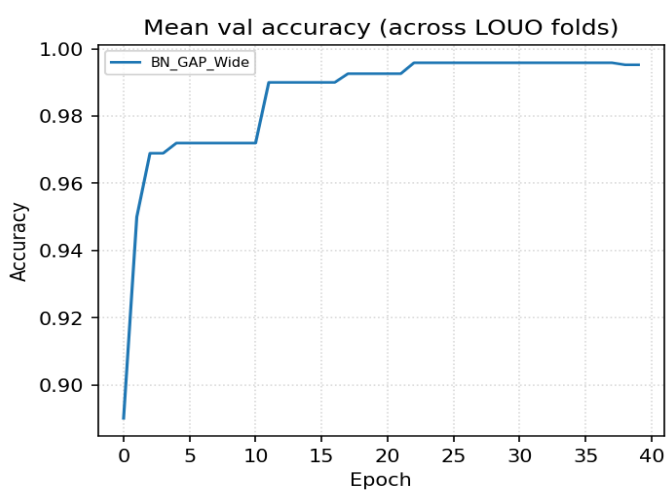
| Variant     | Params  | LOUO mean | LOUO std | Folds | Held-out user acc |
|-------------|---------|-----------|----------|-------|-------------------|
| BN_GAP_Wide | 104,294 | 0.9712    | 0.0711   | 11    | —                 |

Best LOUO mean: <b>BN\_GAP\_Wide</b>

## LOUO accuracy per variant

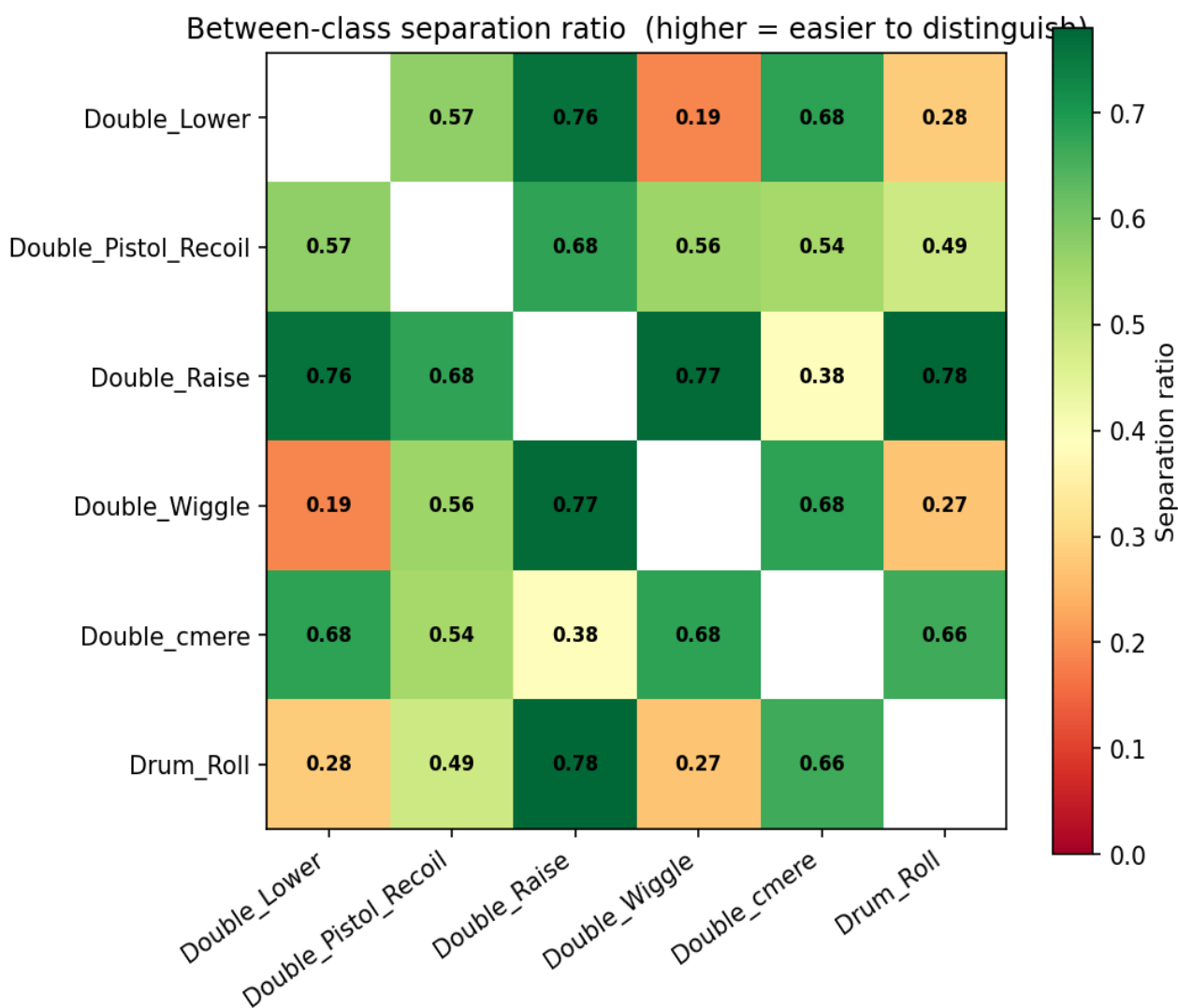
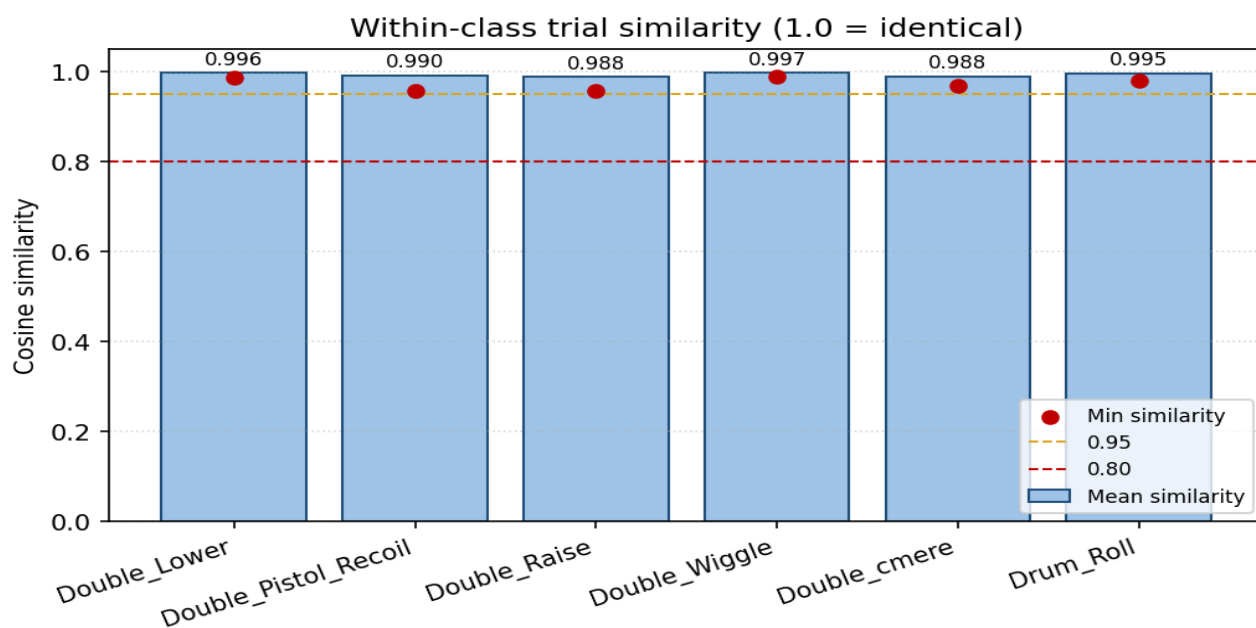


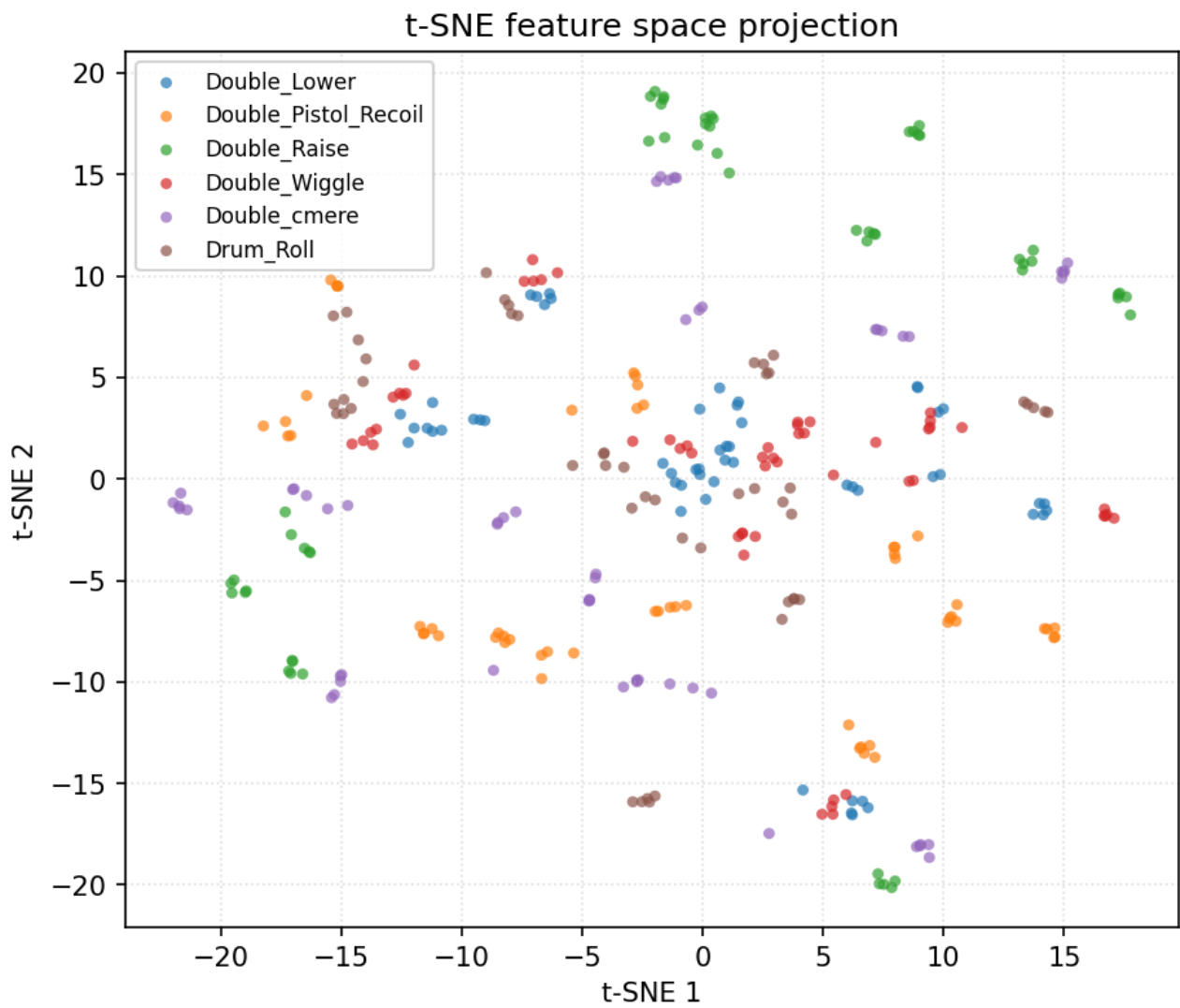
## Mean training curves across LOUO folds



# Data diagnostics — pooled across users

Filtered + resampled trials from every LOUO user (before per-trial normalisation).

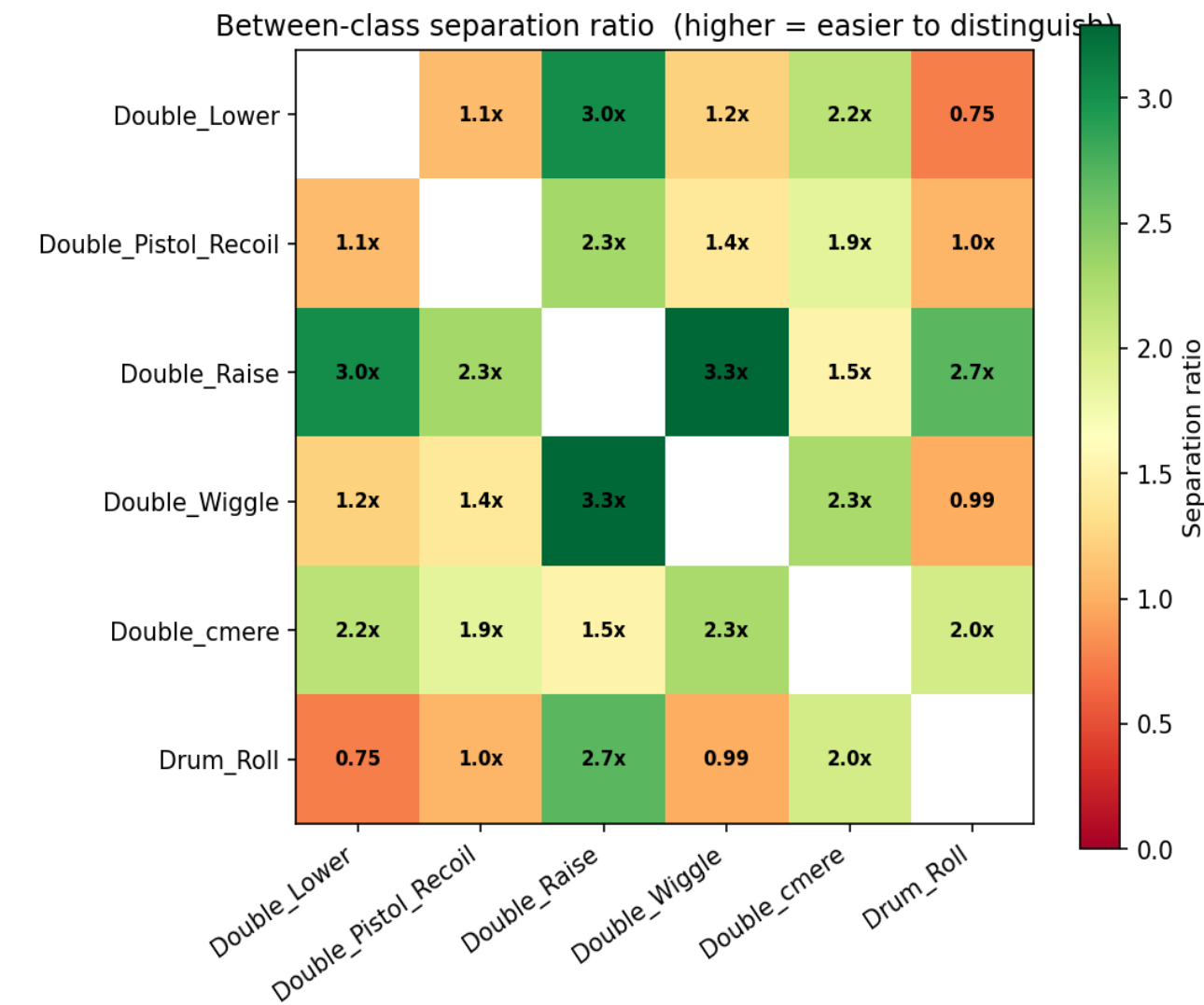
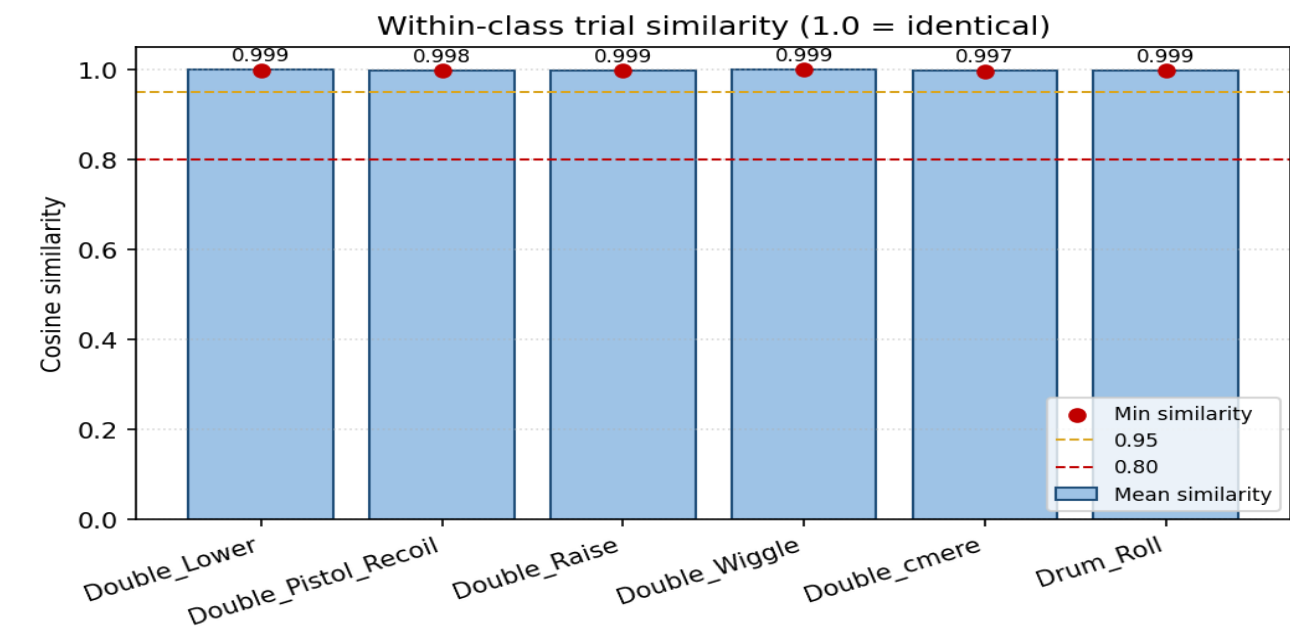




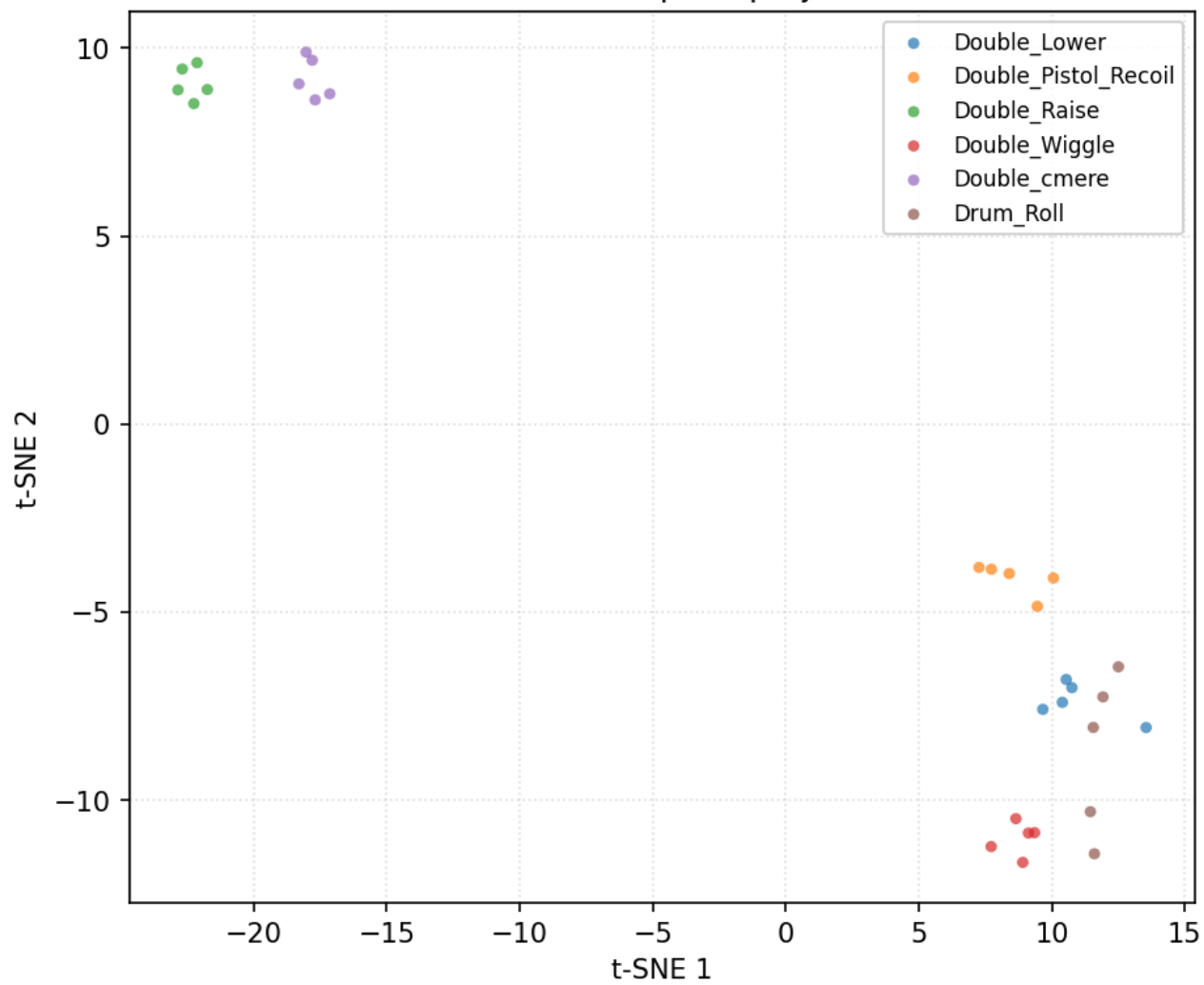
Reading guide: within-class similarity > 0.95 = very consistent; < 0.80 = high variance. Separation ratio > 100x = trivially separable, 20-100x = easy, 5-20x = moderate, < 5x = hard (expect confusion).

# Data diagnostics — user 11\_Mansh

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

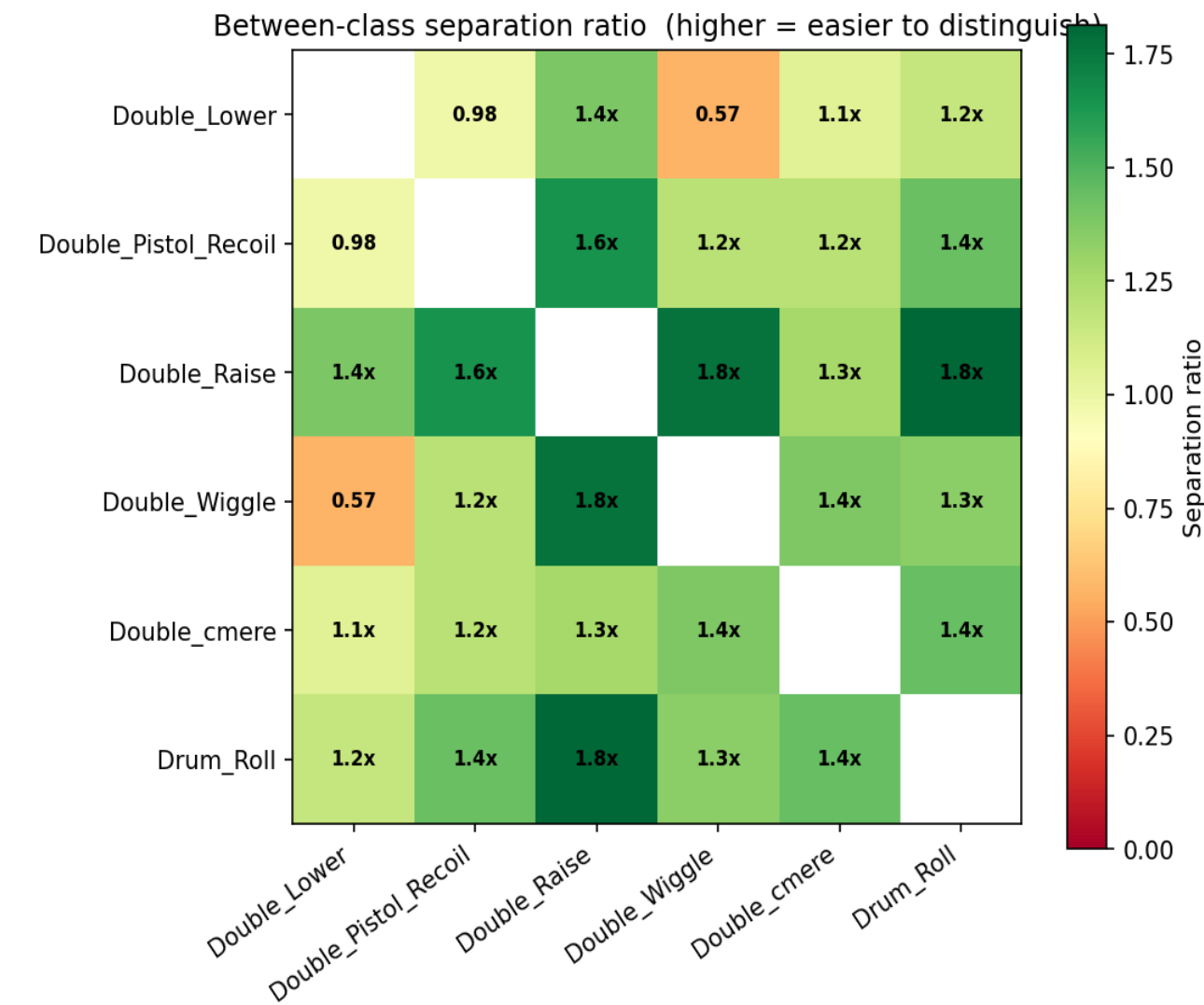
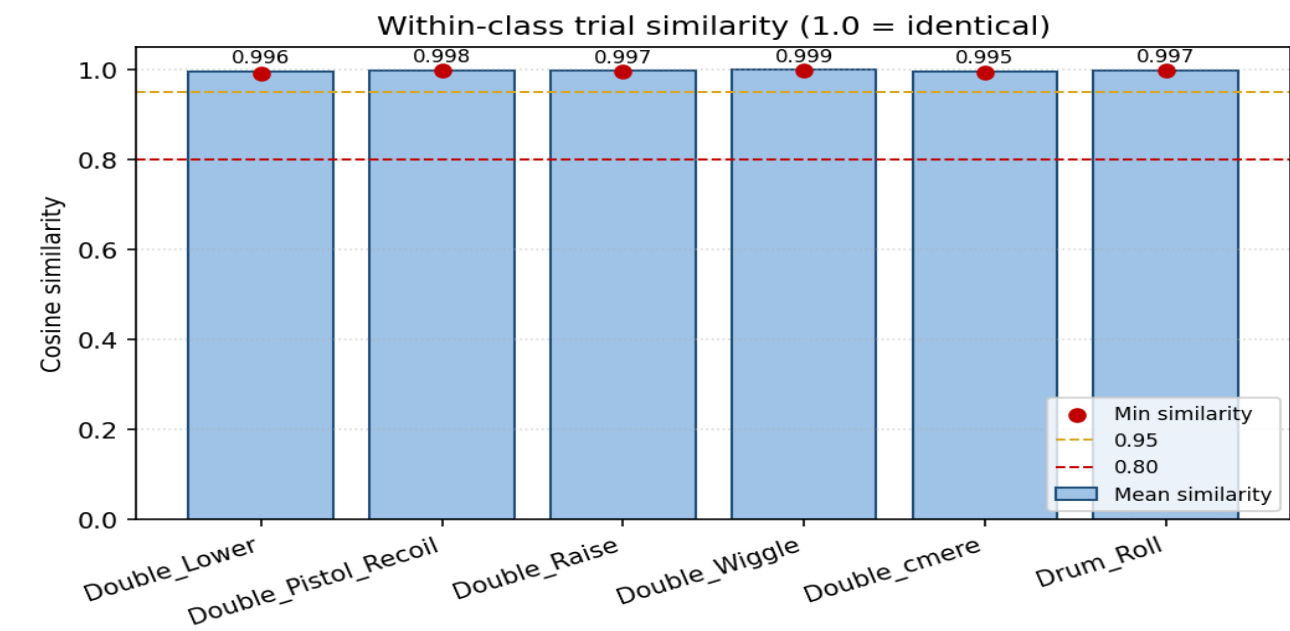


t-SNE feature space projection

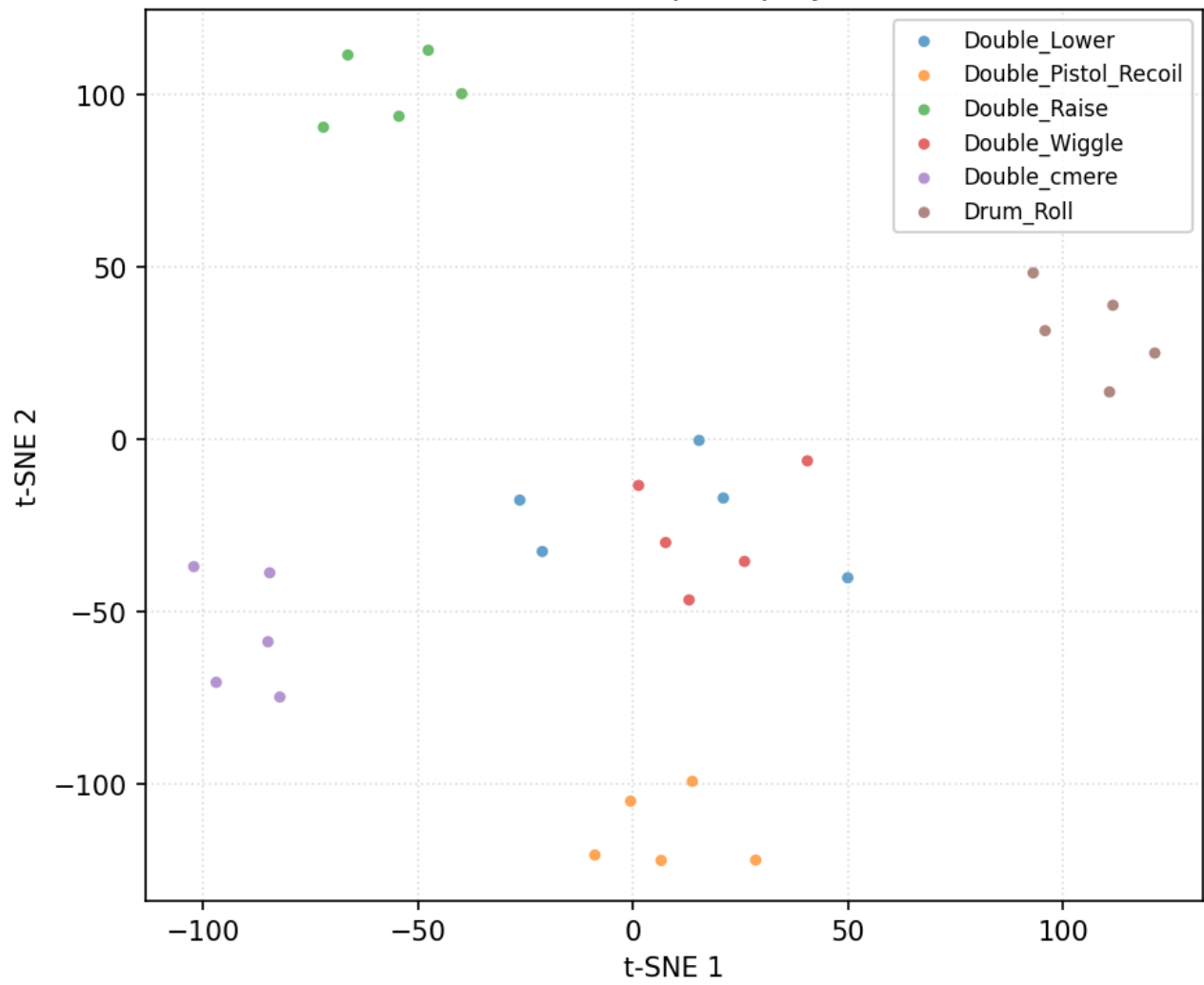


# Data diagnostics — user 12\_Marcus

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).



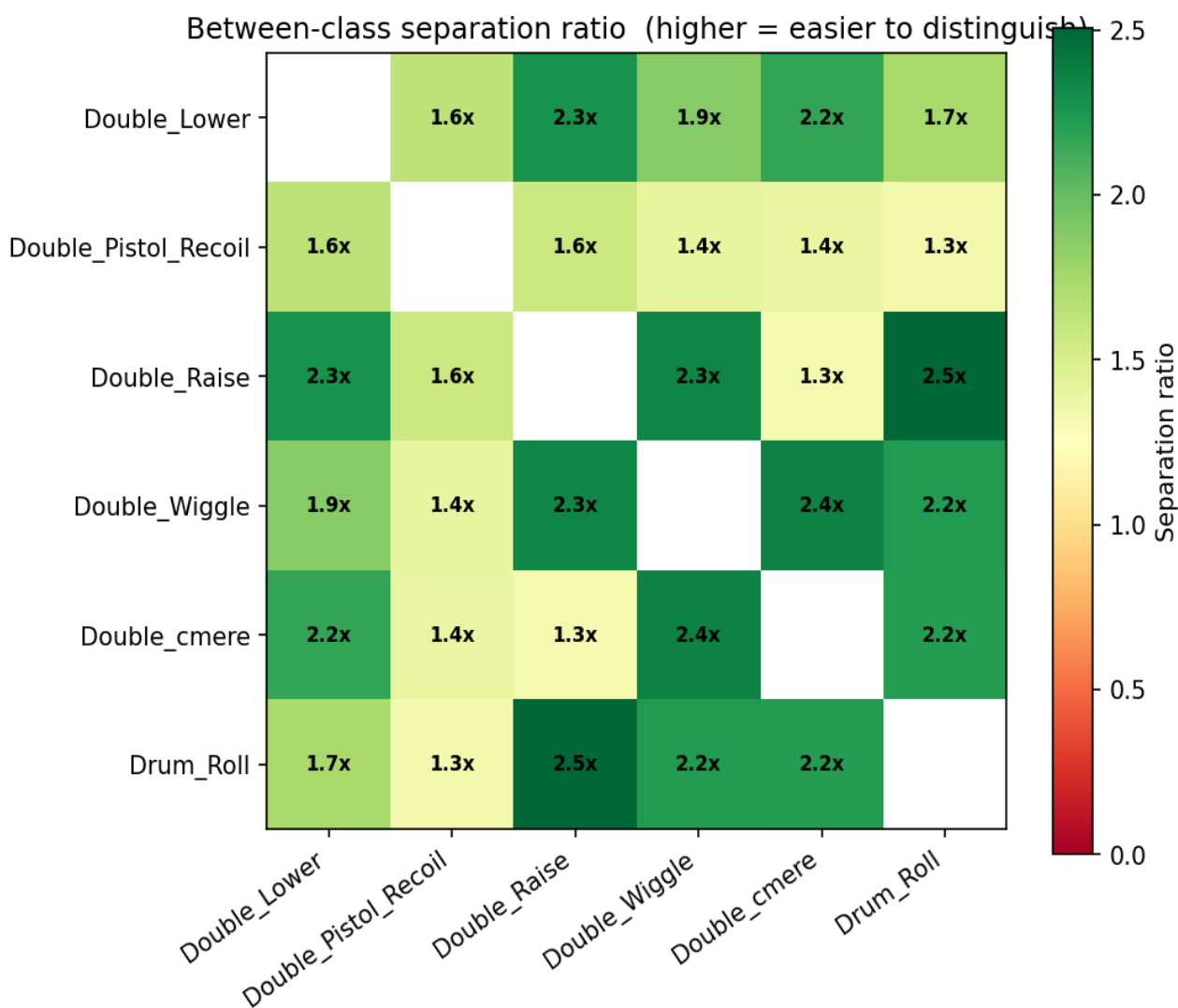
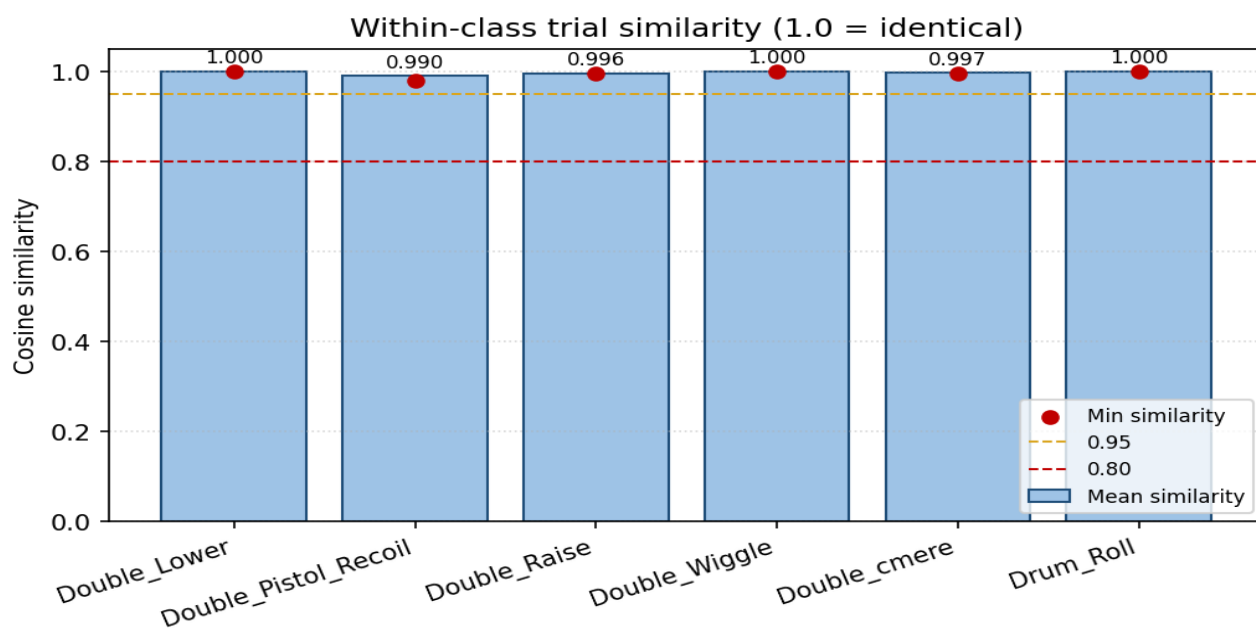
t-SNE feature space projection



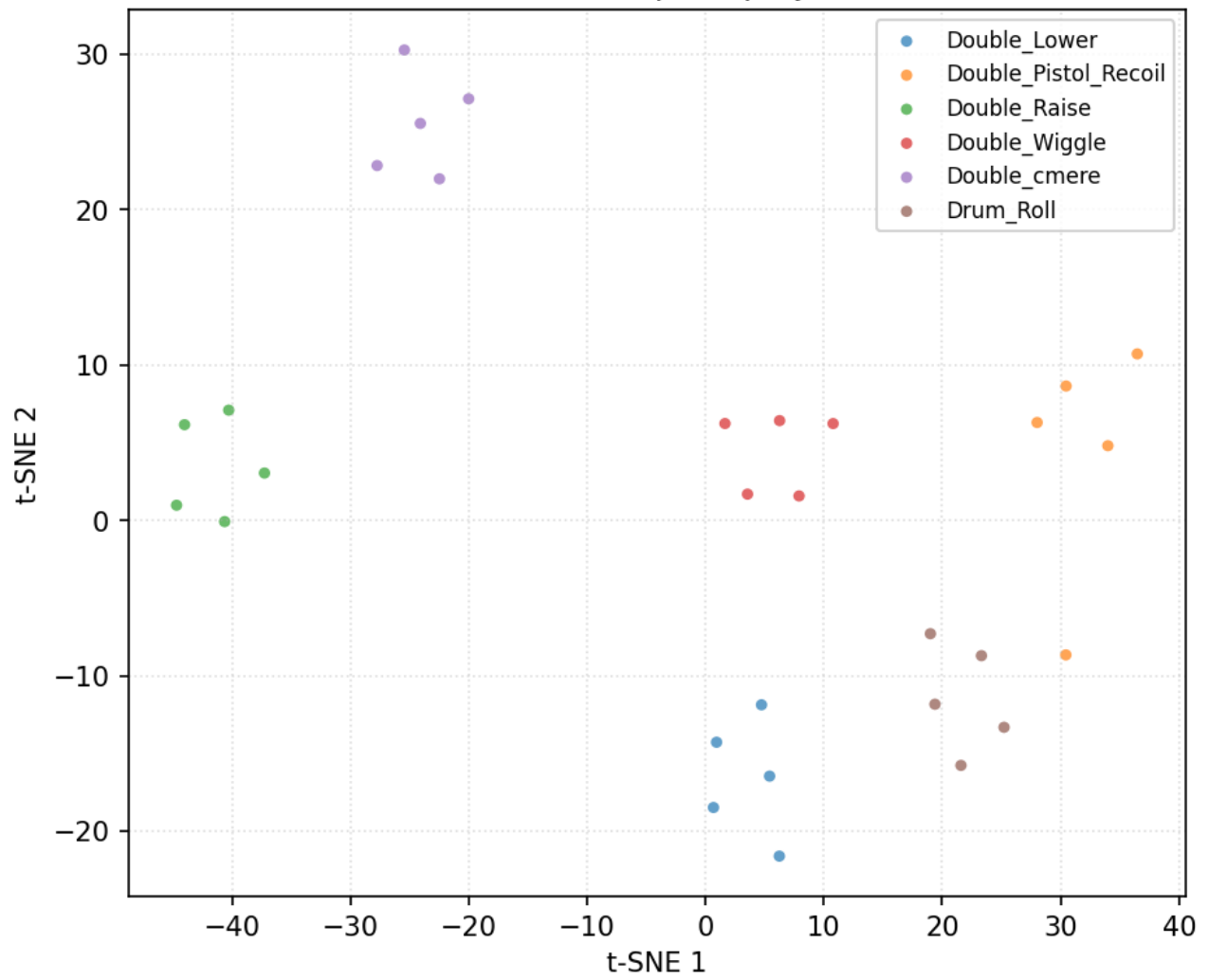


# Data diagnostics — user 14\_StephenV2

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

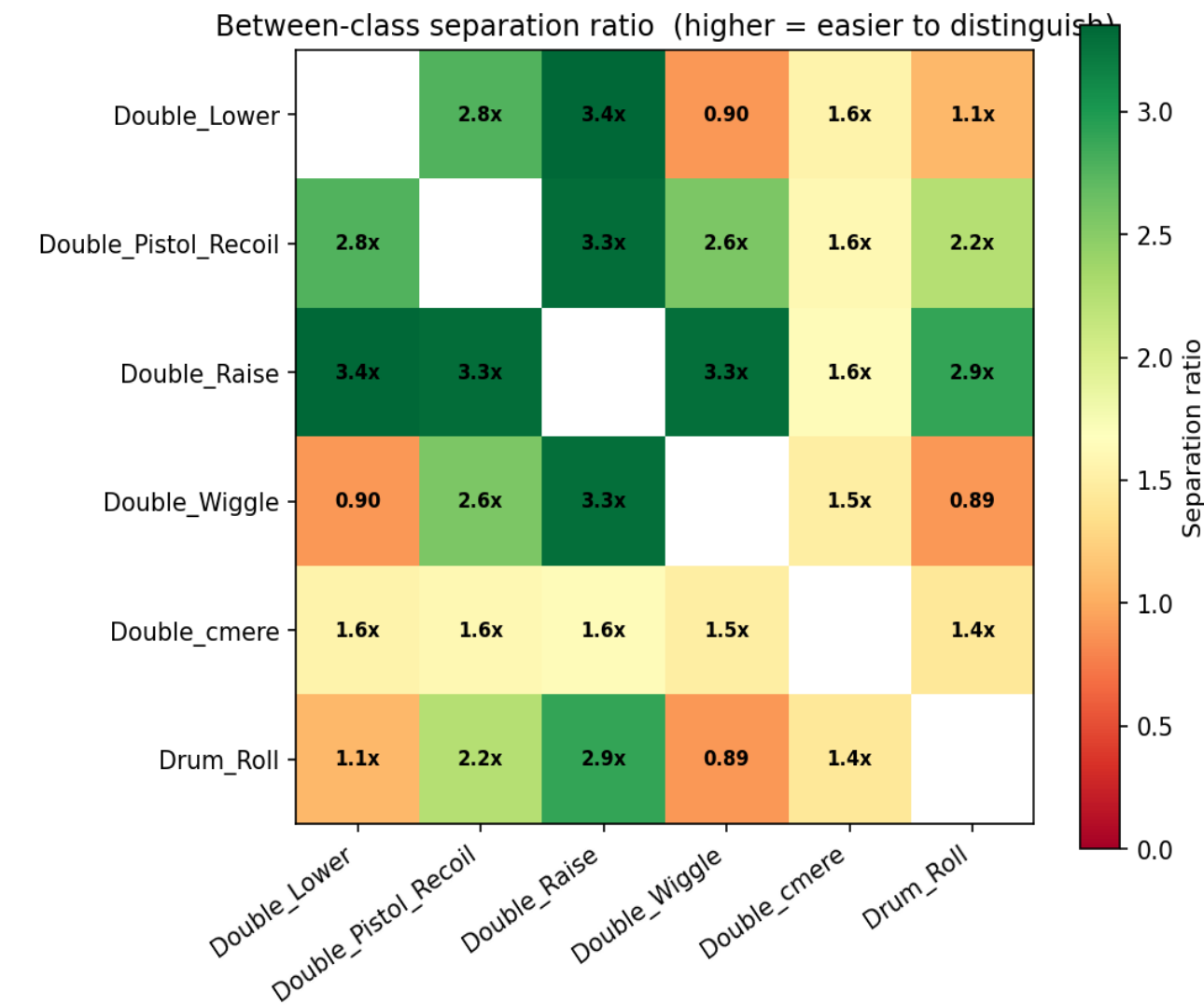
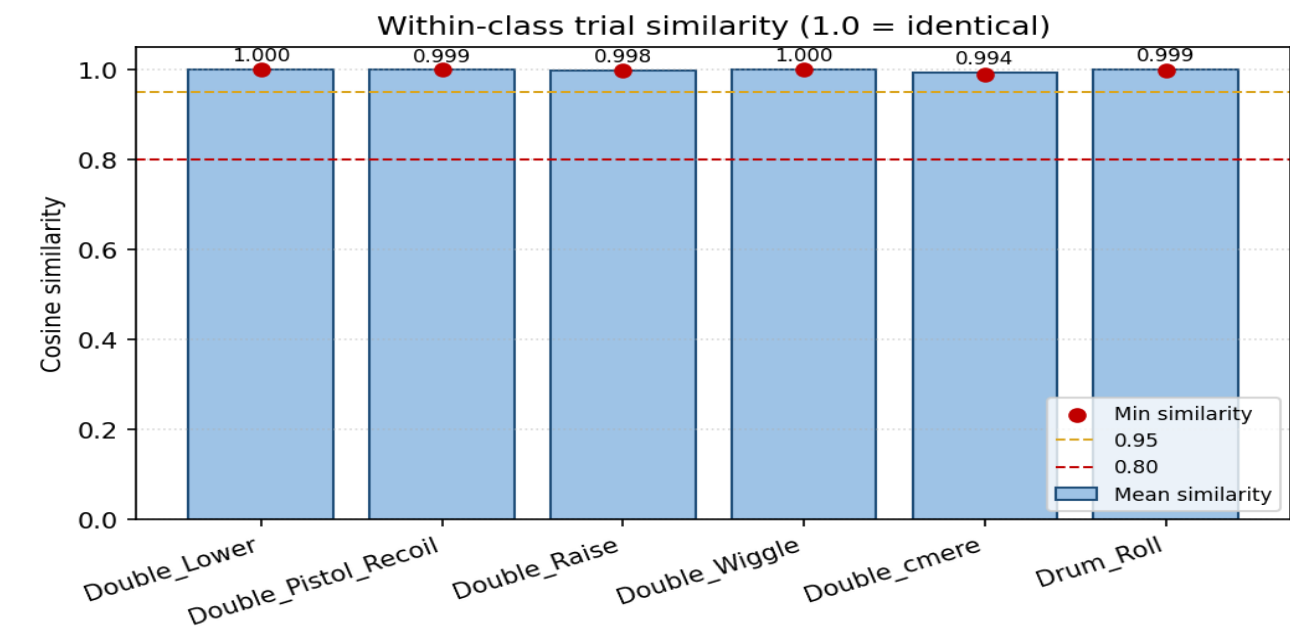


t-SNE feature space projection

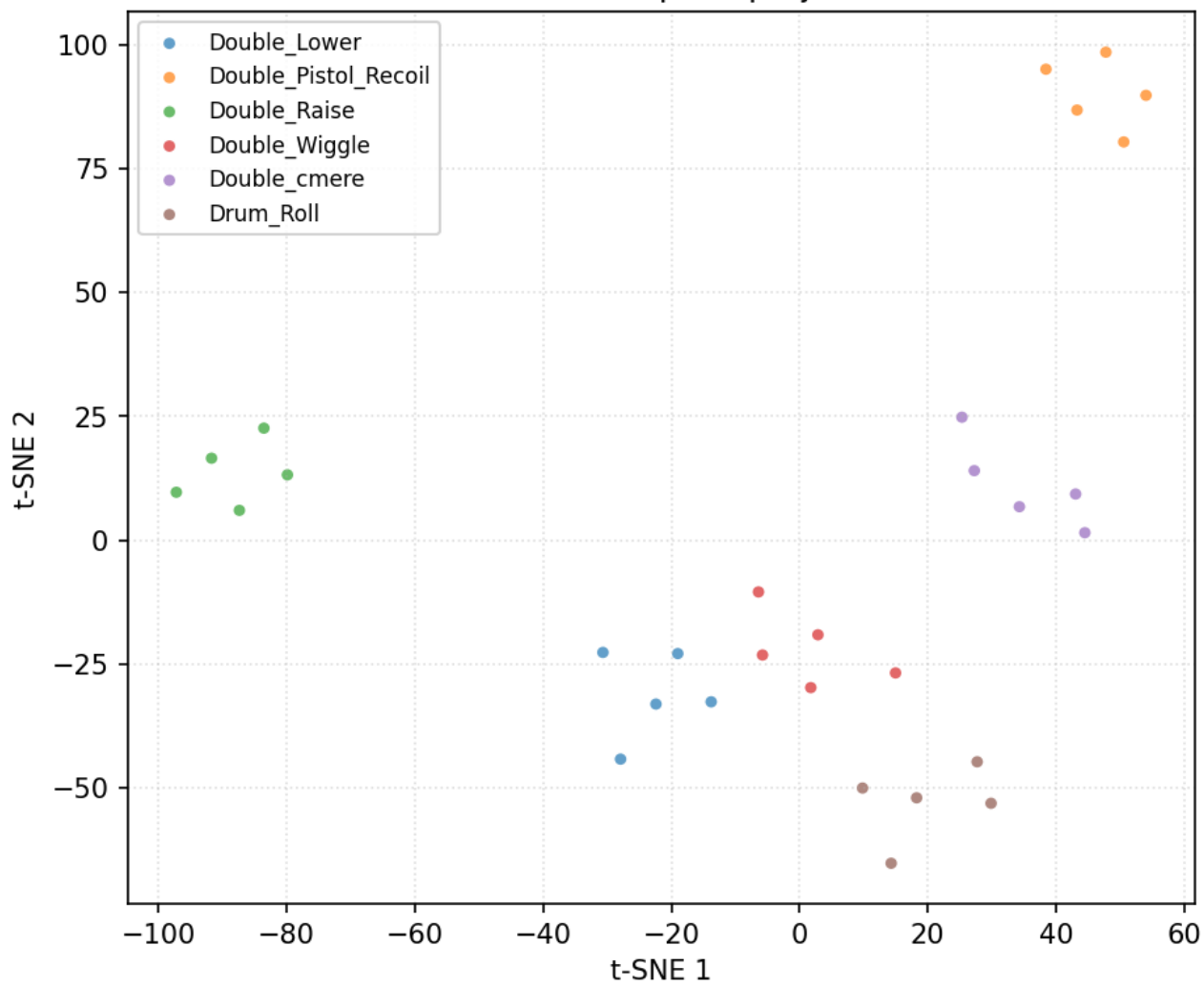


# Data diagnostics — user 17\_Raquel

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

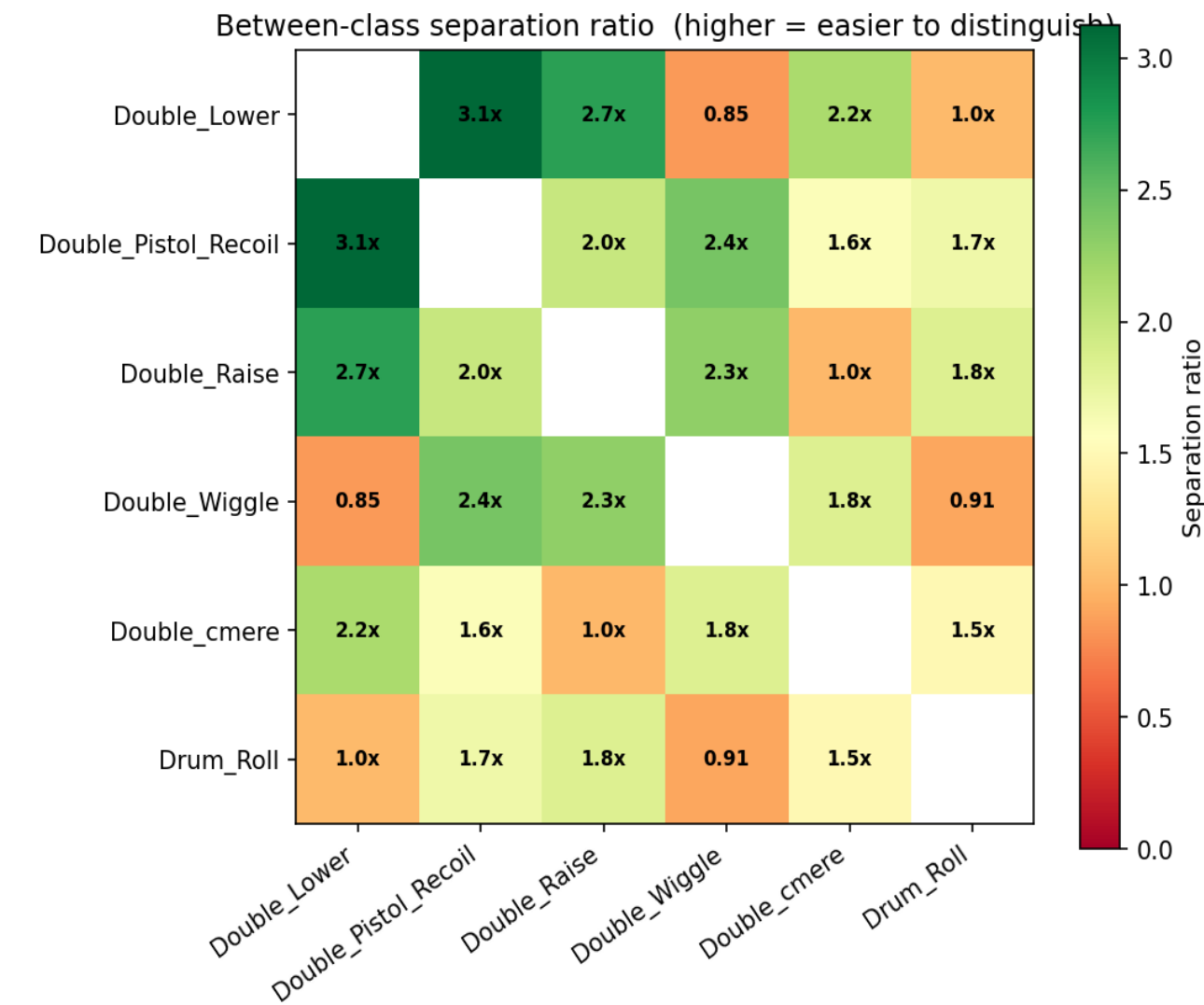
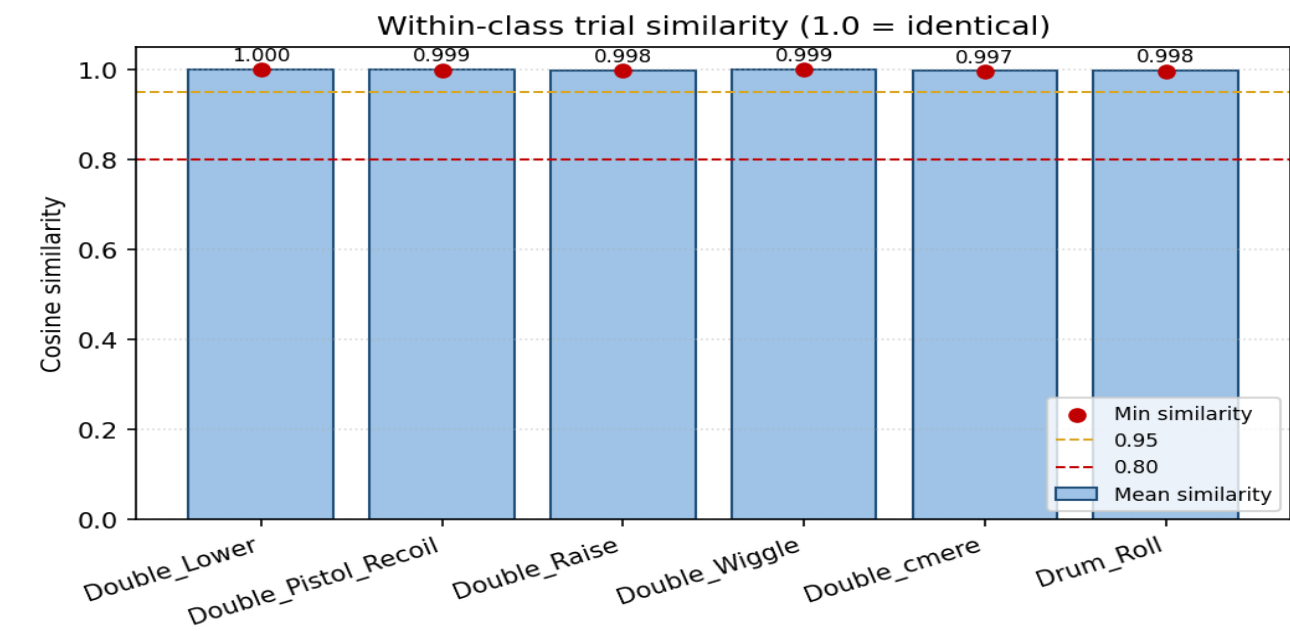


t-SNE feature space projection

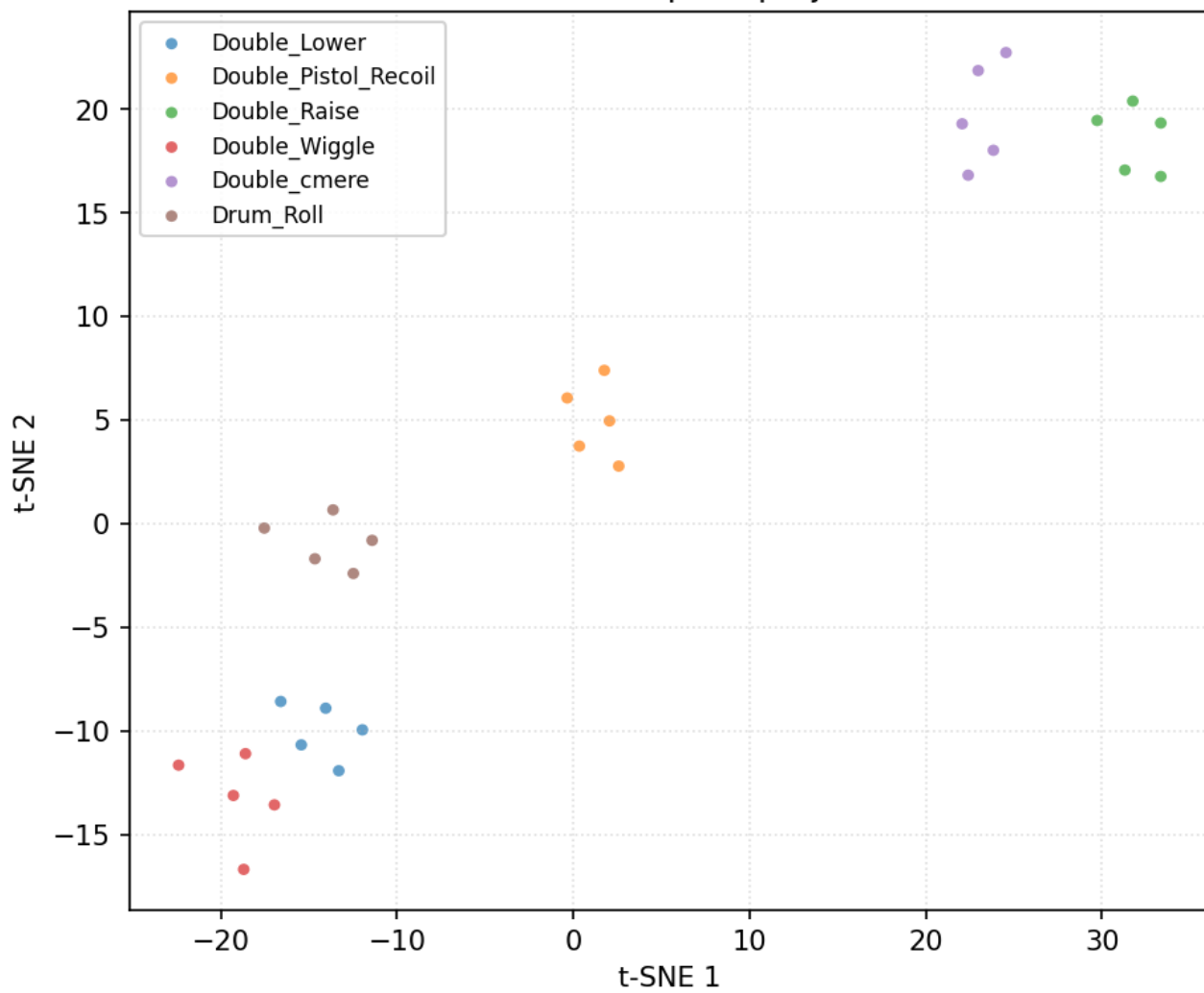


# Data diagnostics — user 18\_Bridgette

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

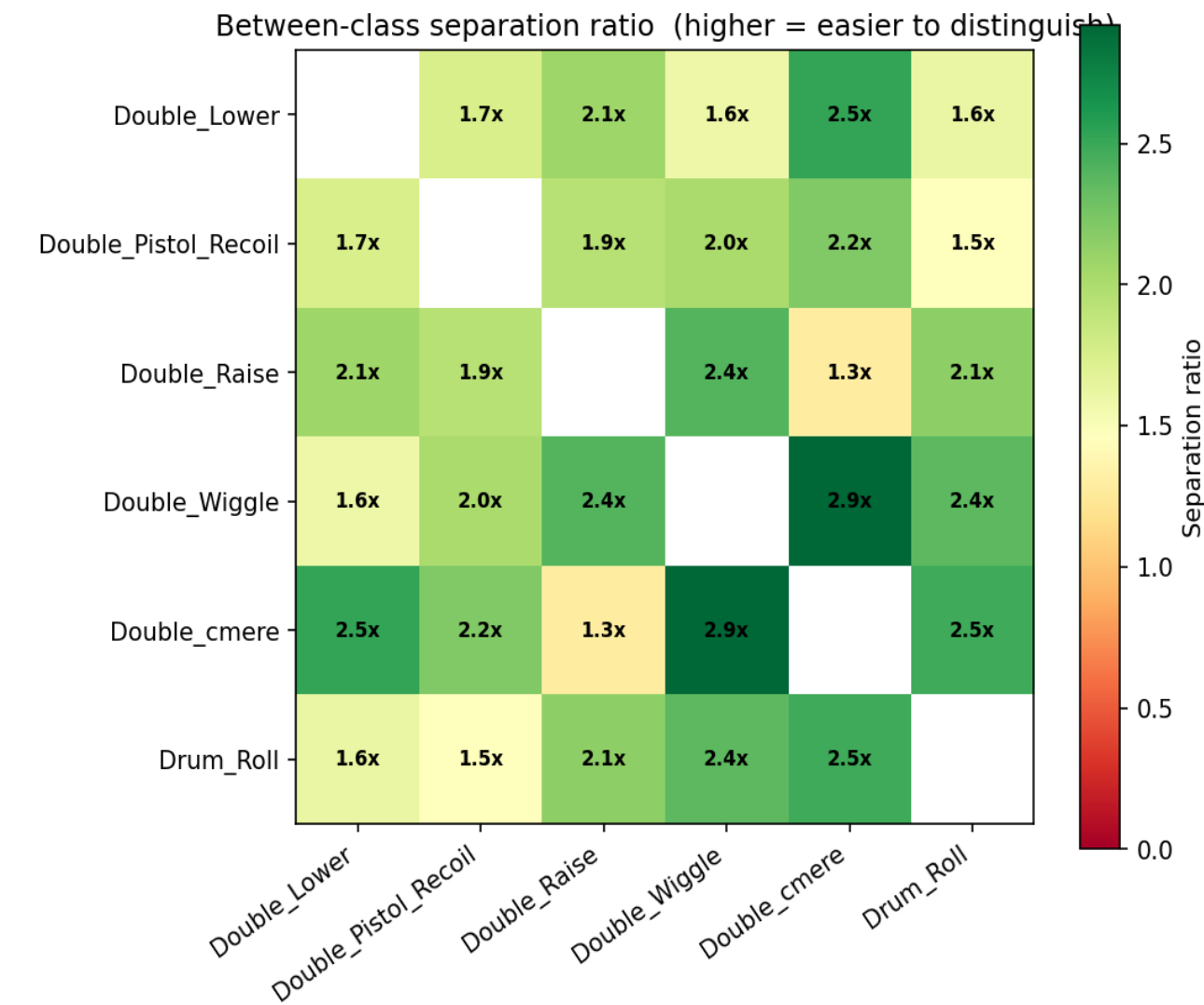
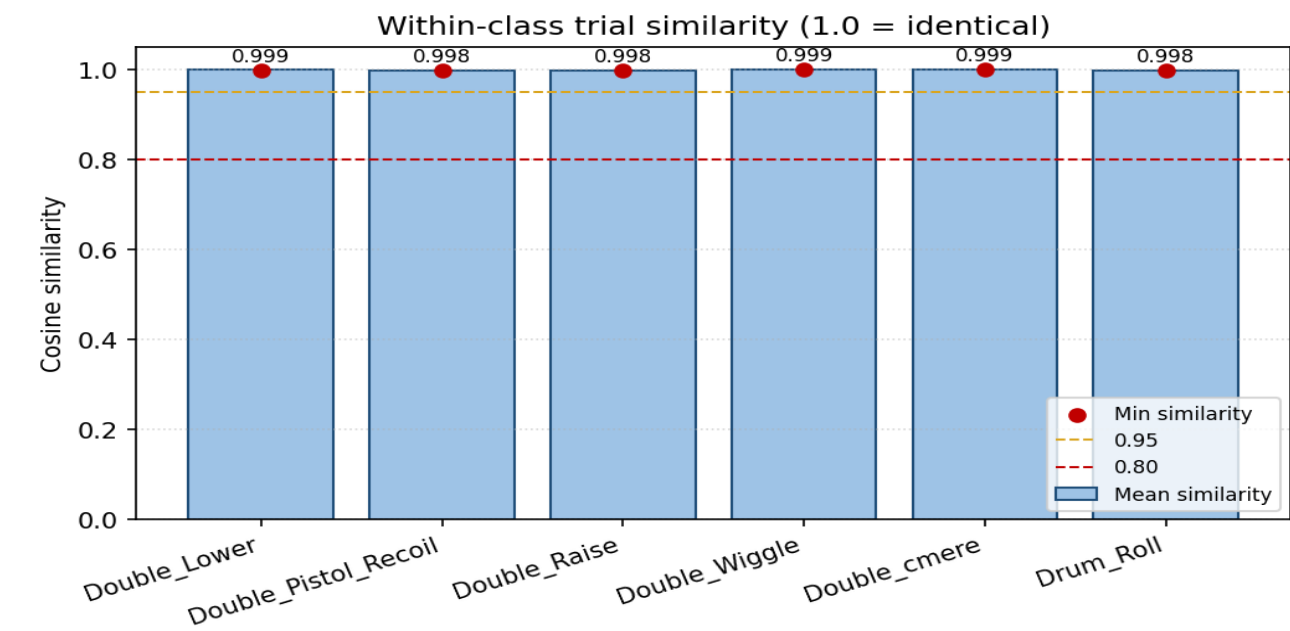


t-SNE feature space projection

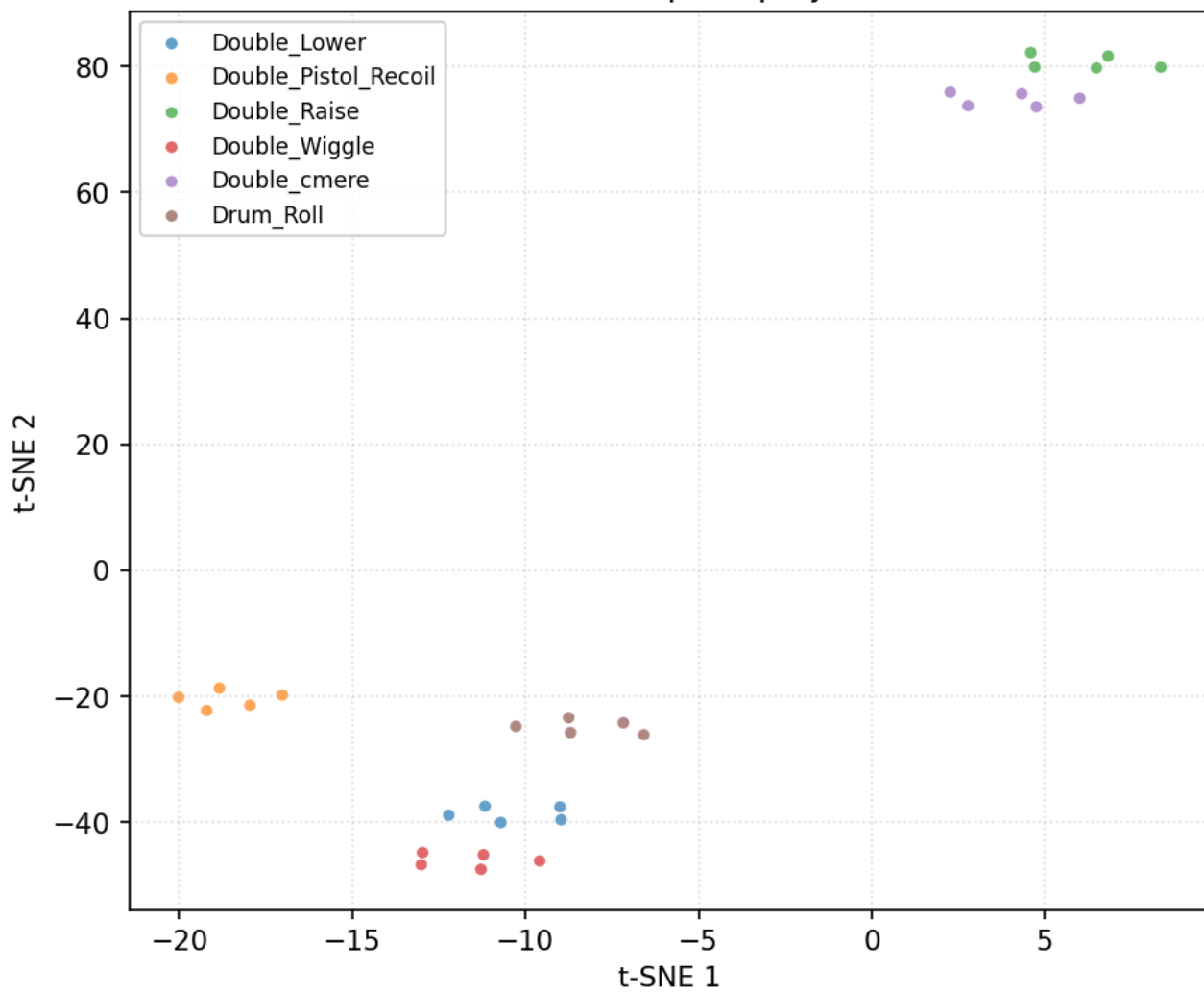


# Data diagnostics — user 1\_Alán

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).



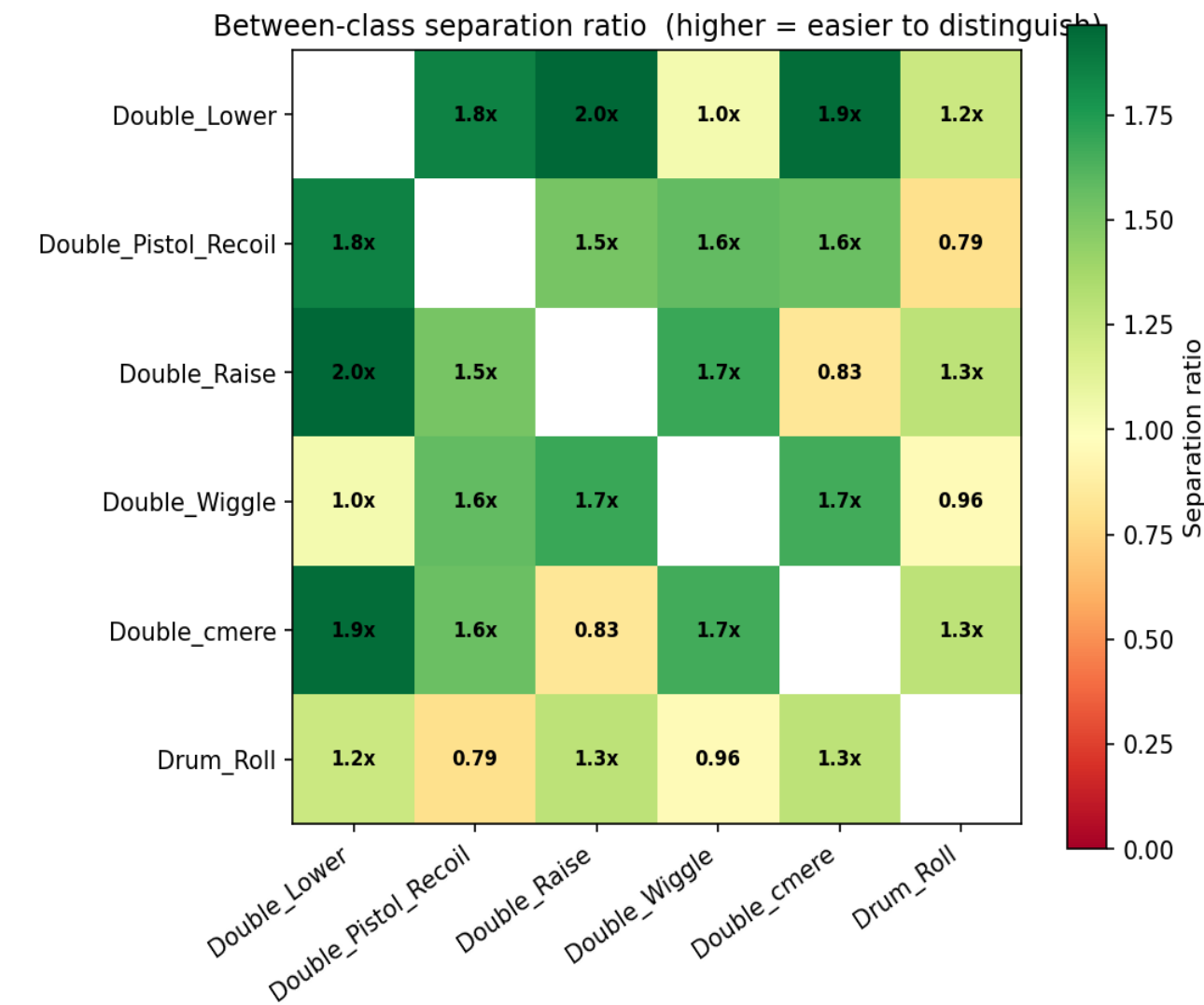
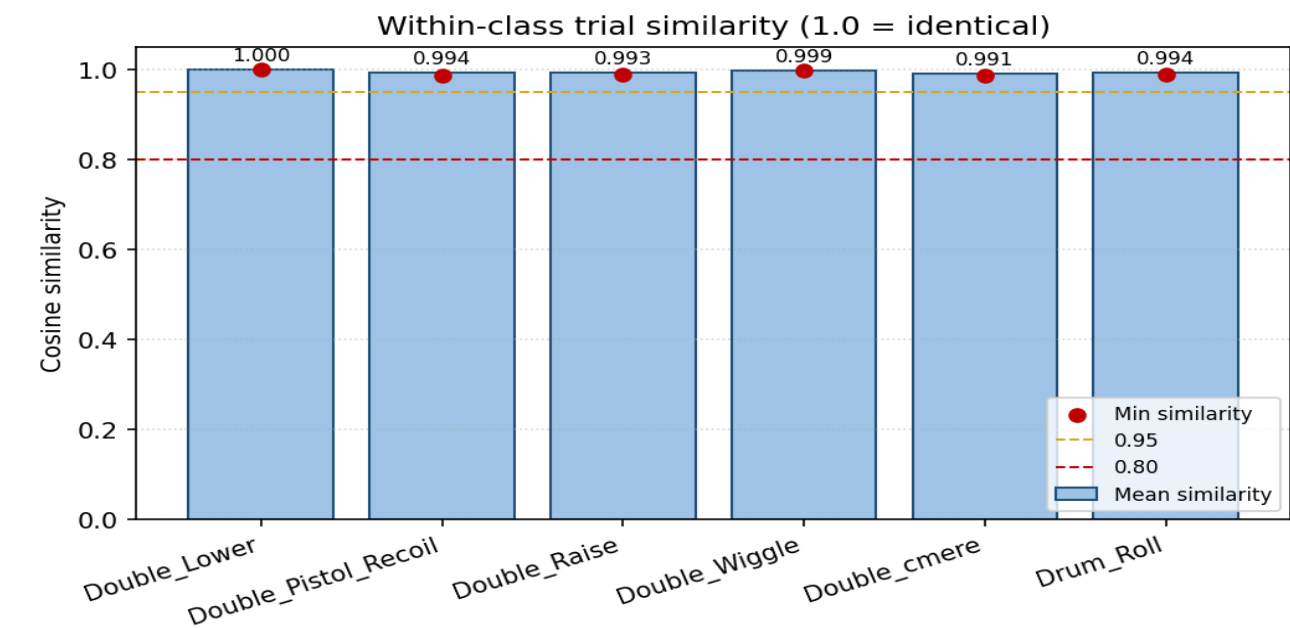
t-SNE feature space projection



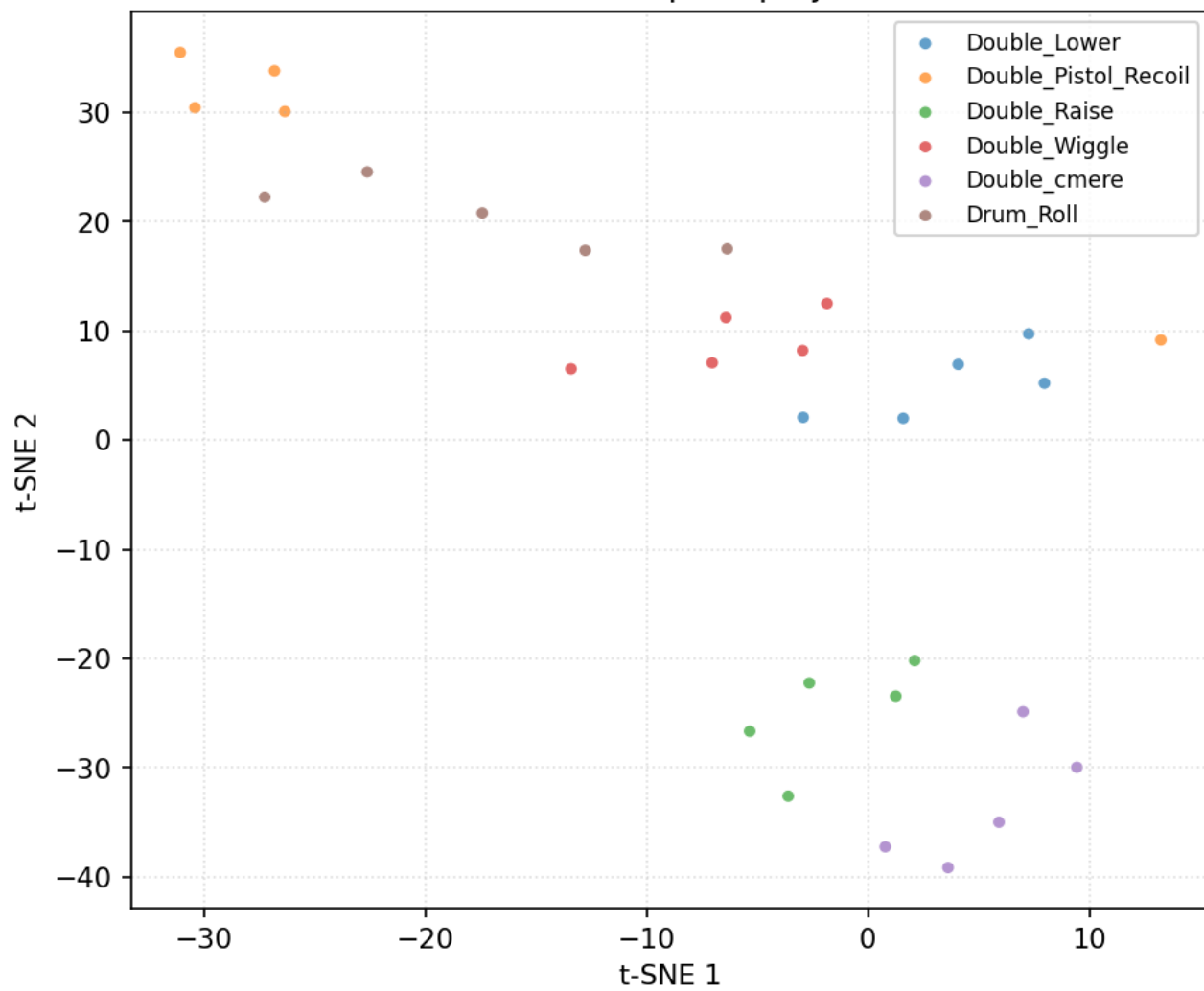


# Data diagnostics — user 2\_Alex

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

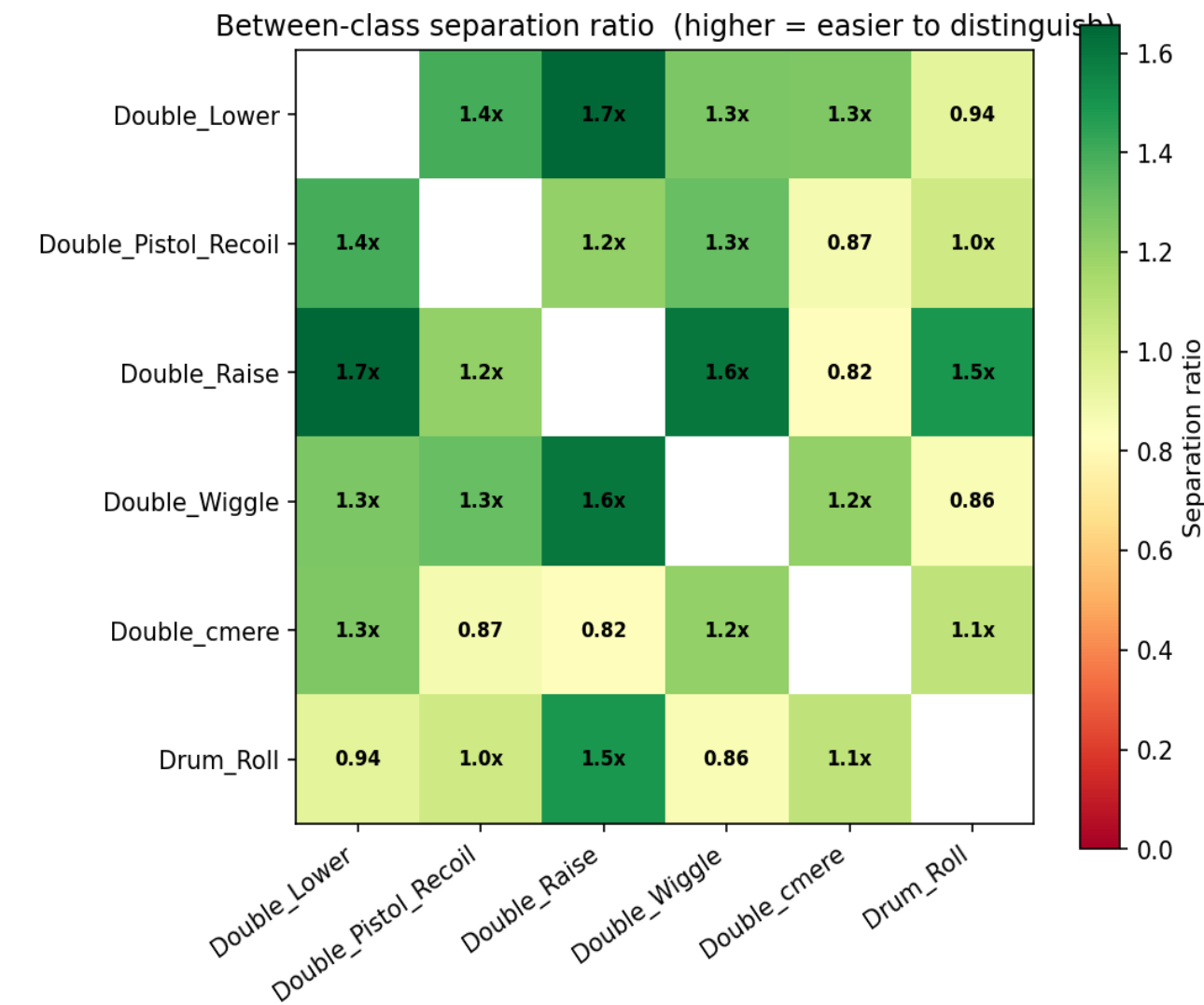
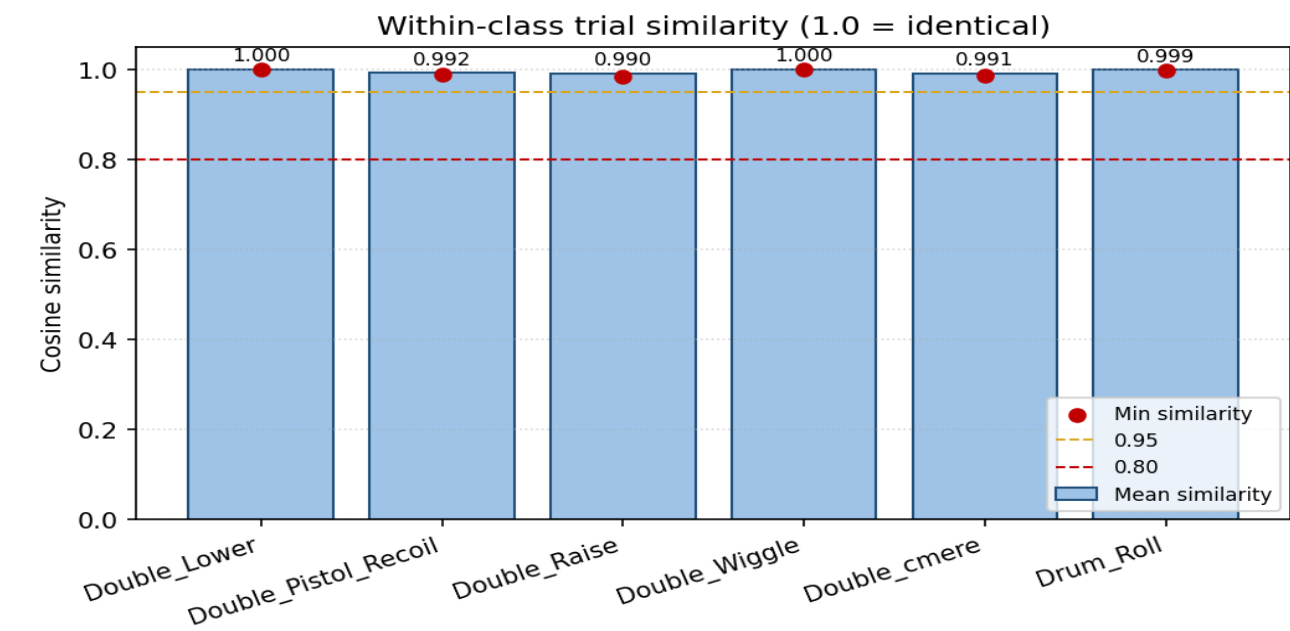


t-SNE feature space projection

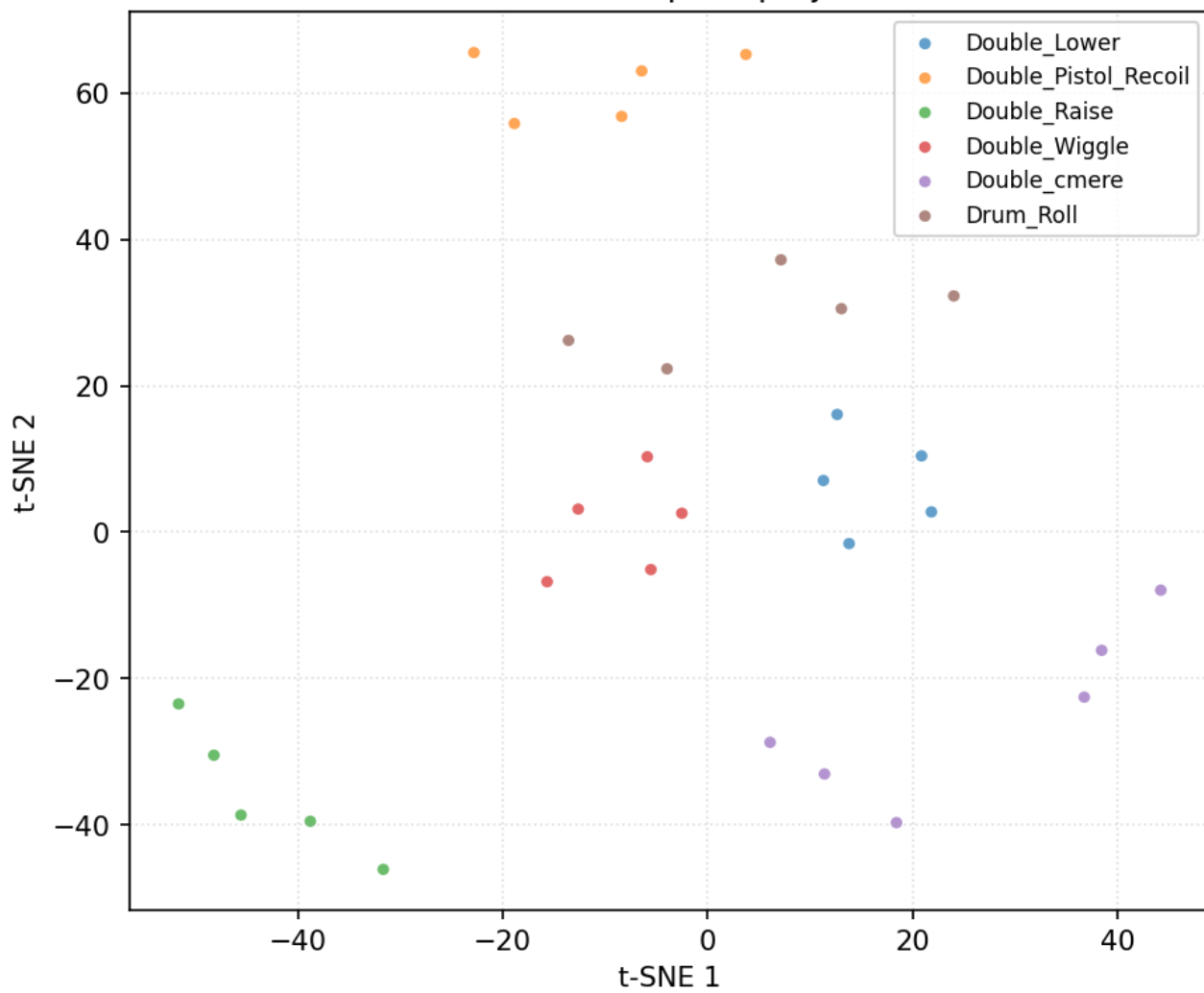


# Data diagnostics — user 3\_Anghad

31 trials, 6 classes (filtered + resampled, before per-trial normalisation).

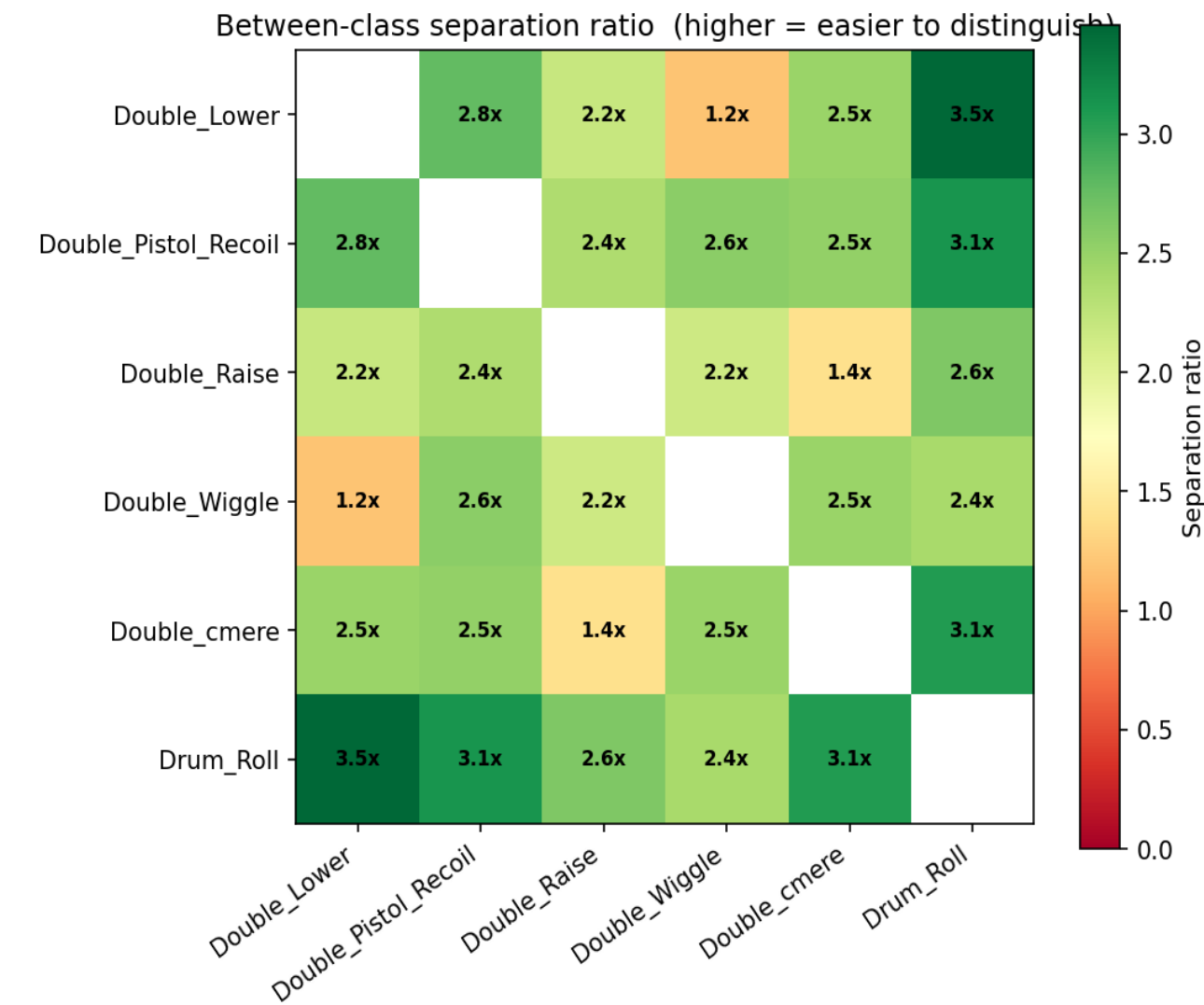
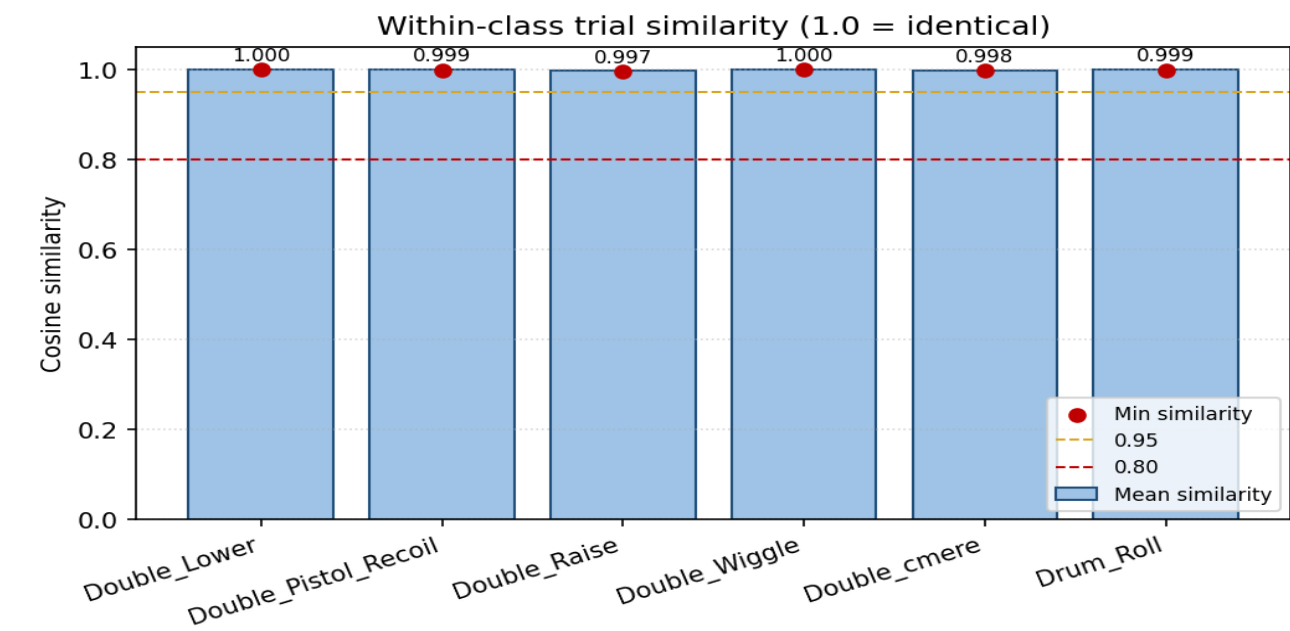


t-SNE feature space projection

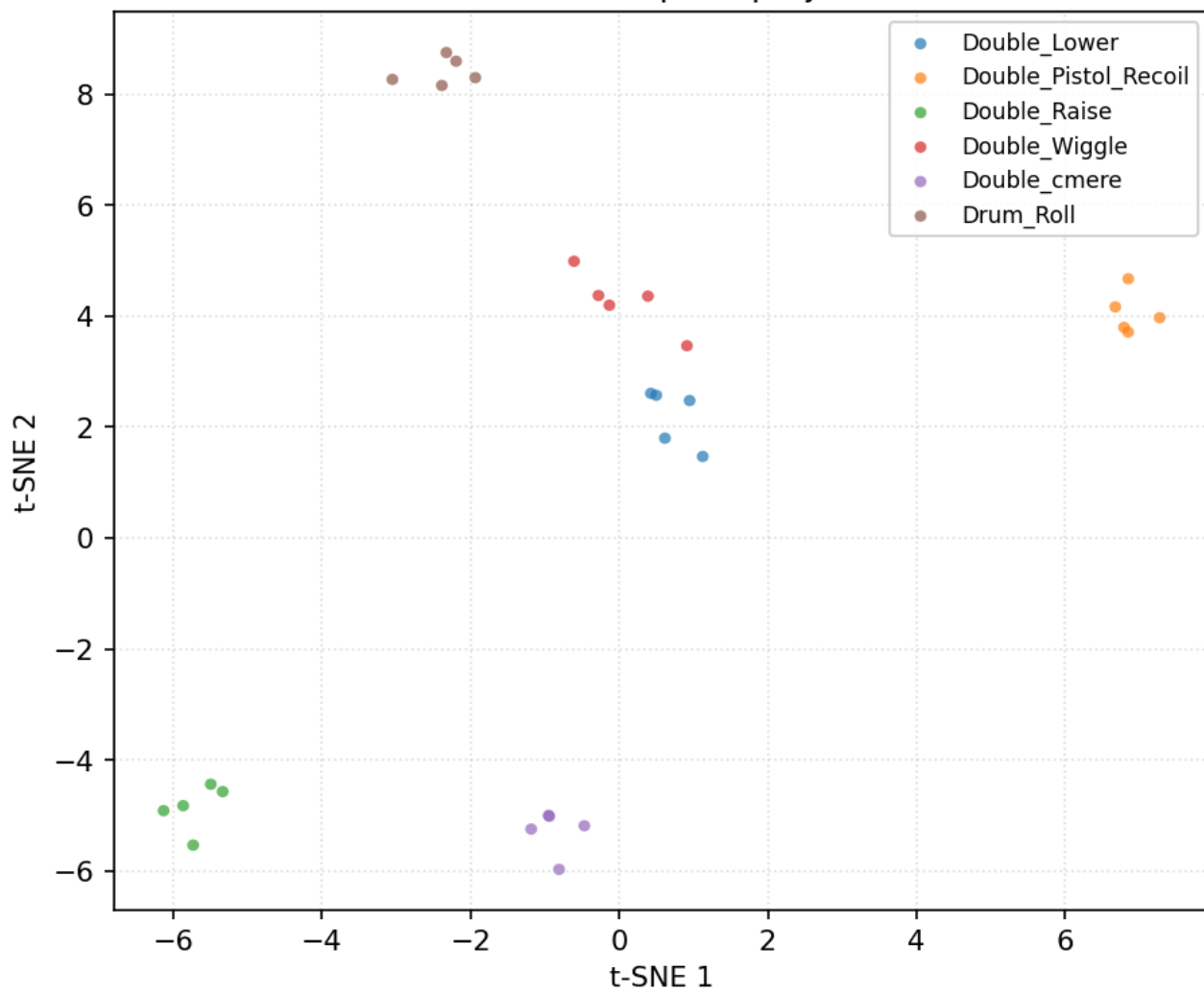


# Data diagnostics — user 4\_Daniel

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

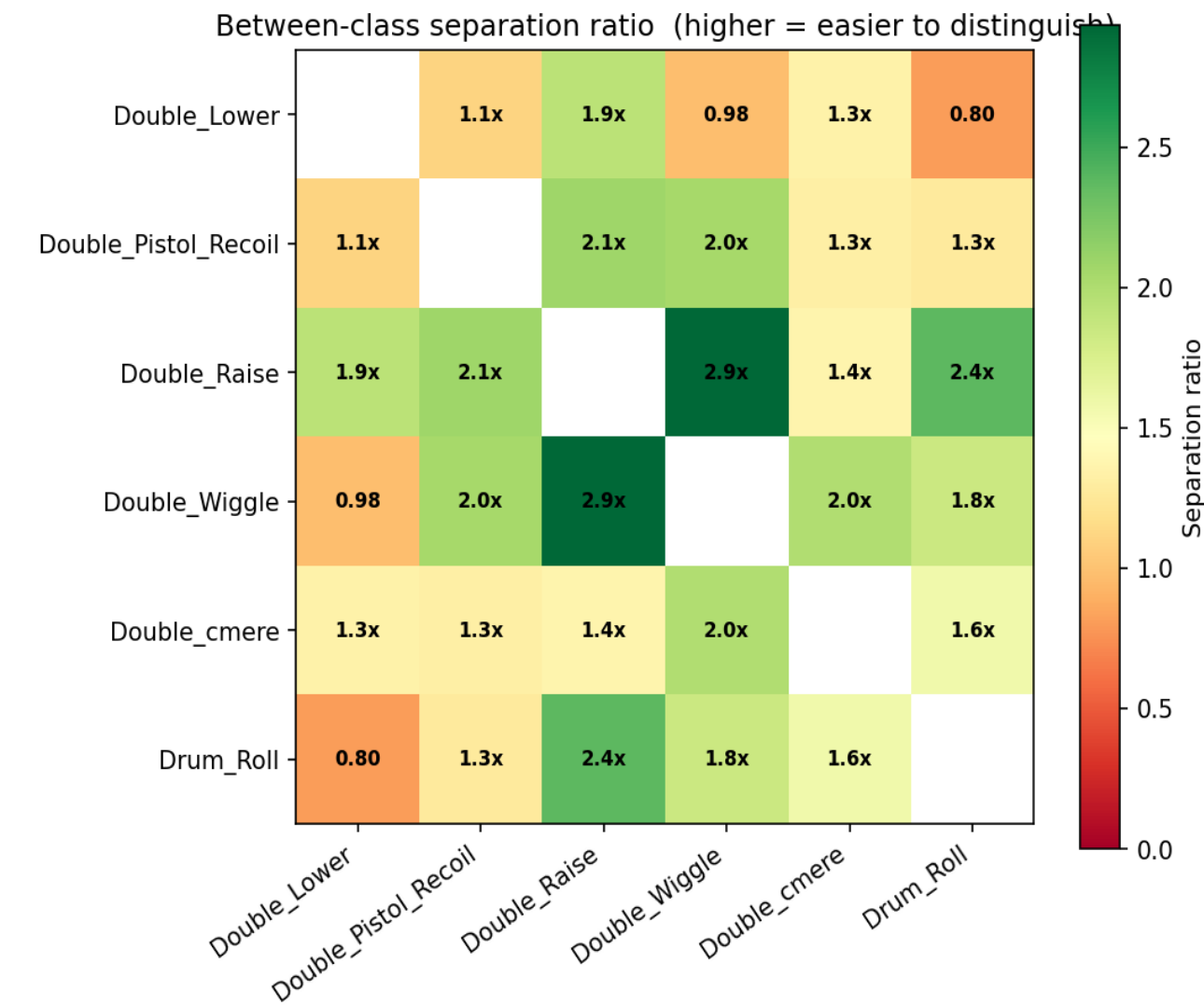
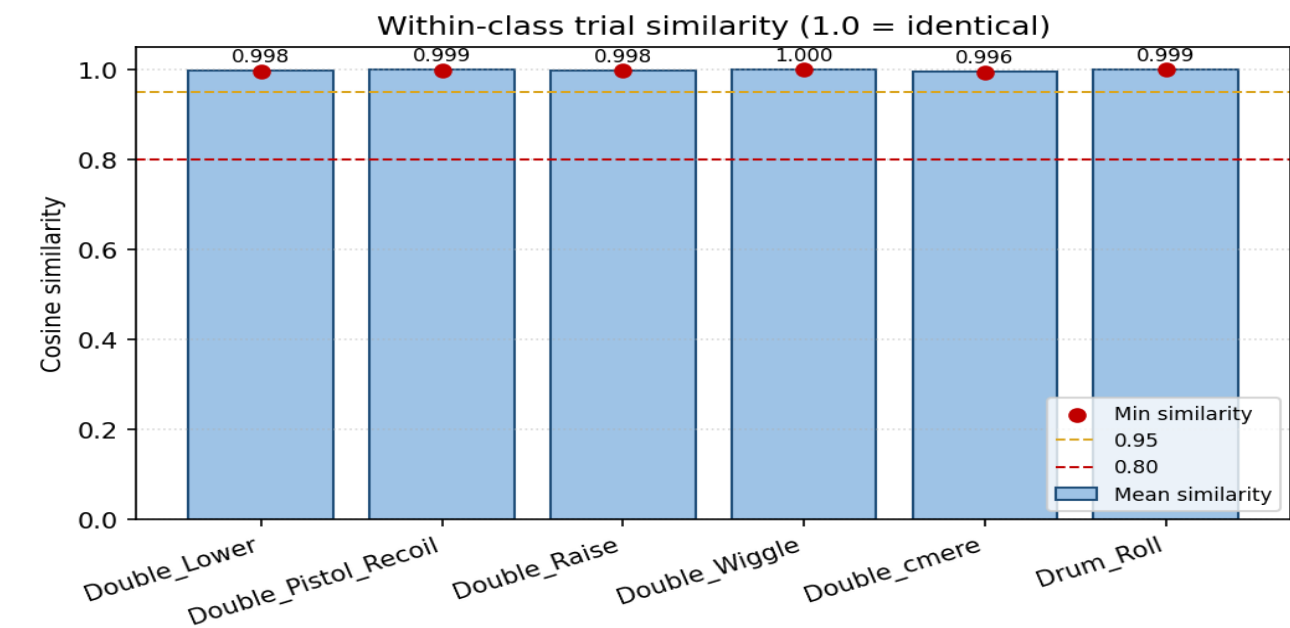


t-SNE feature space projection

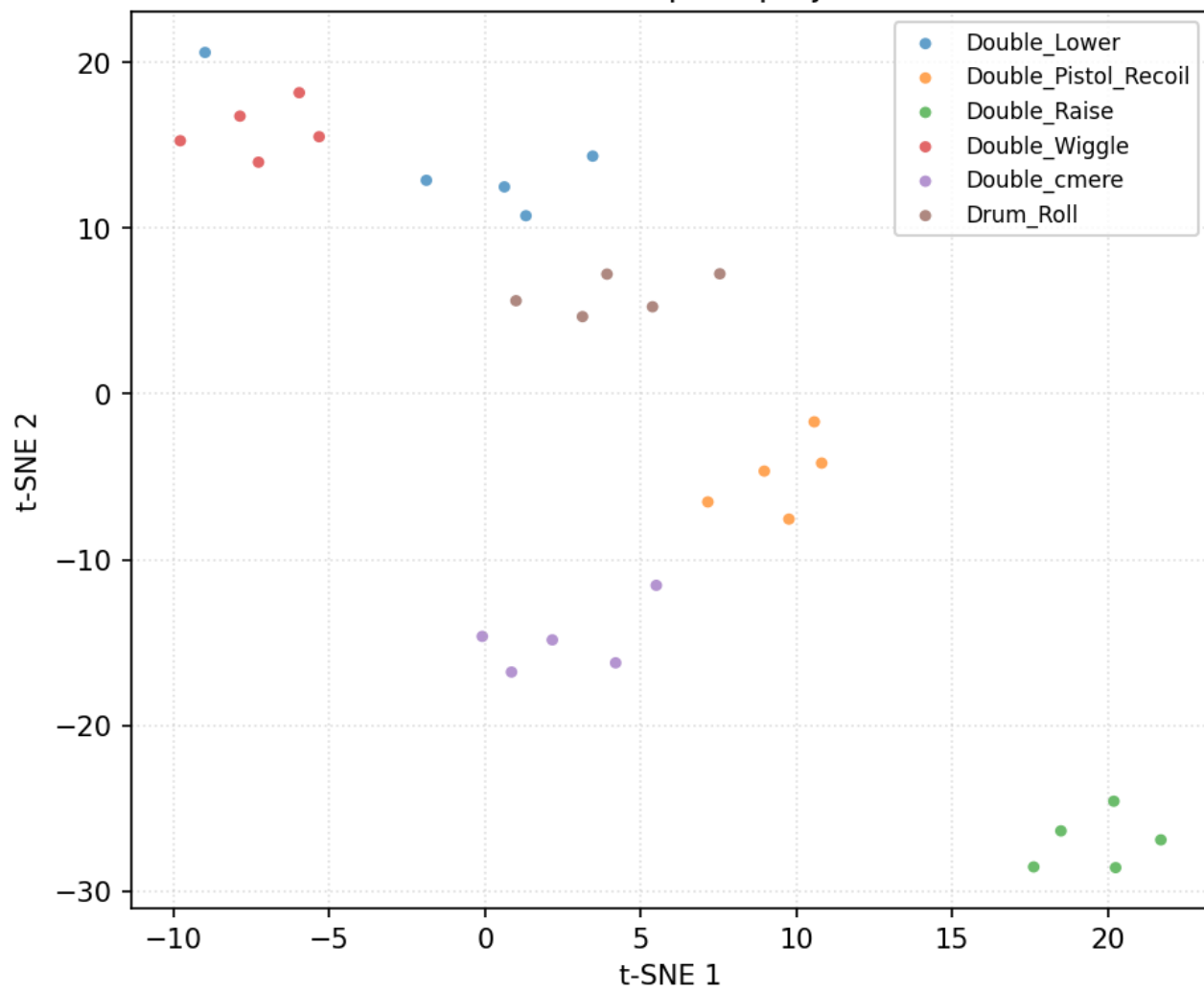


# Data diagnostics — user 6\_Harry

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).



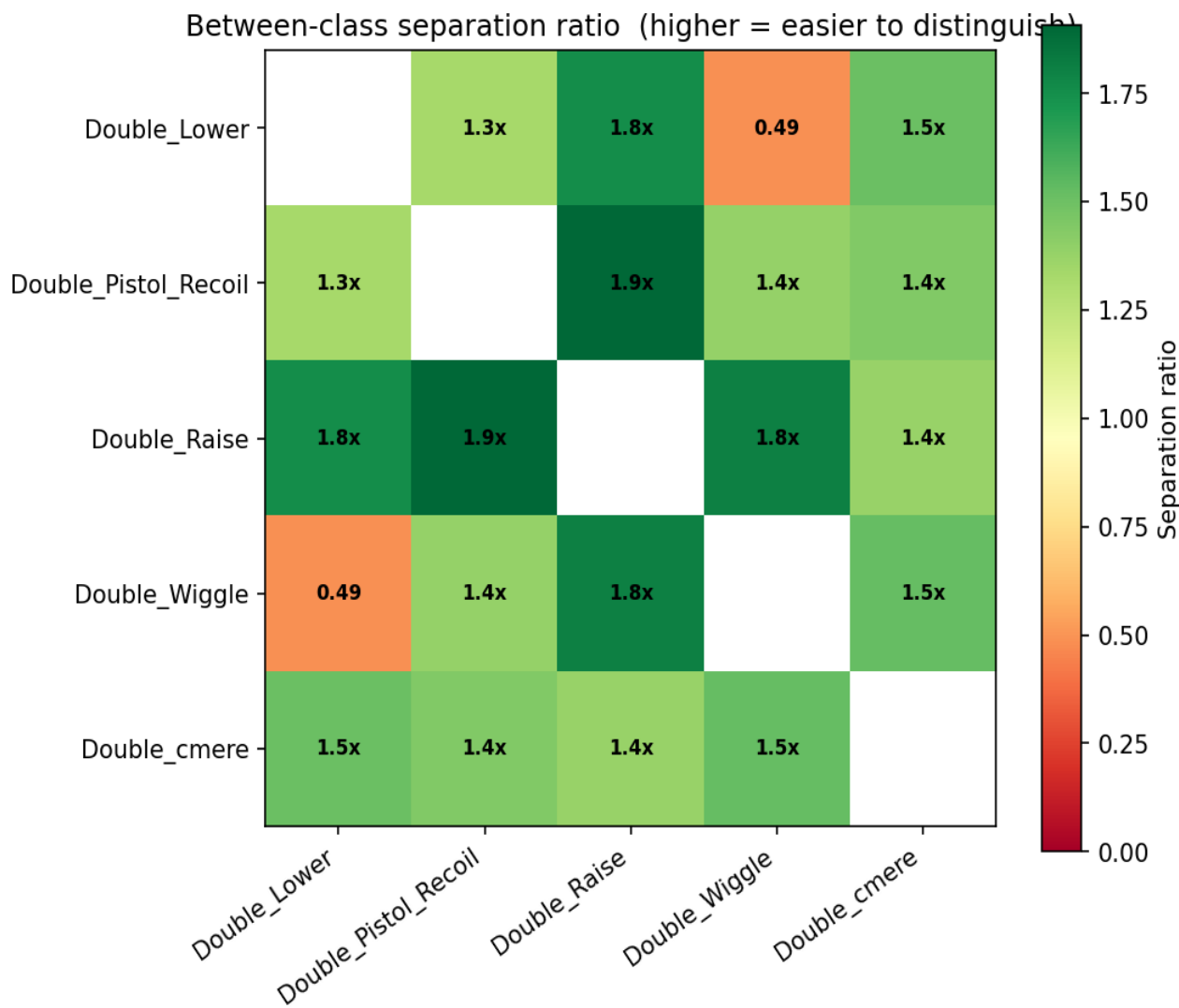
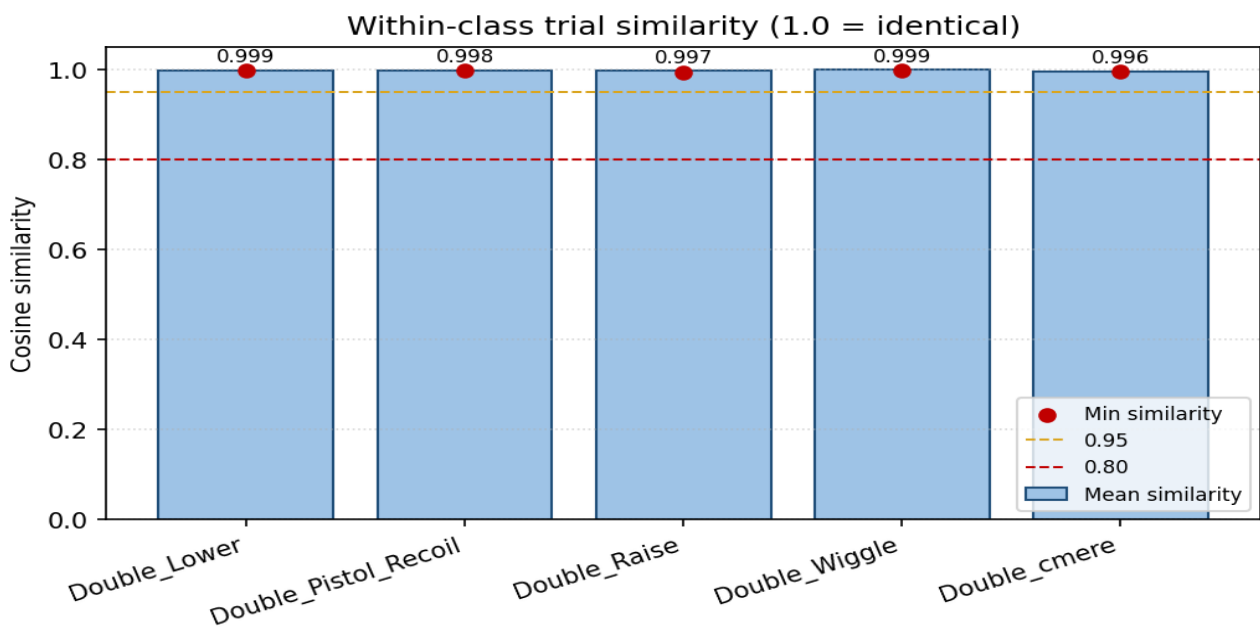
t-SNE feature space projection



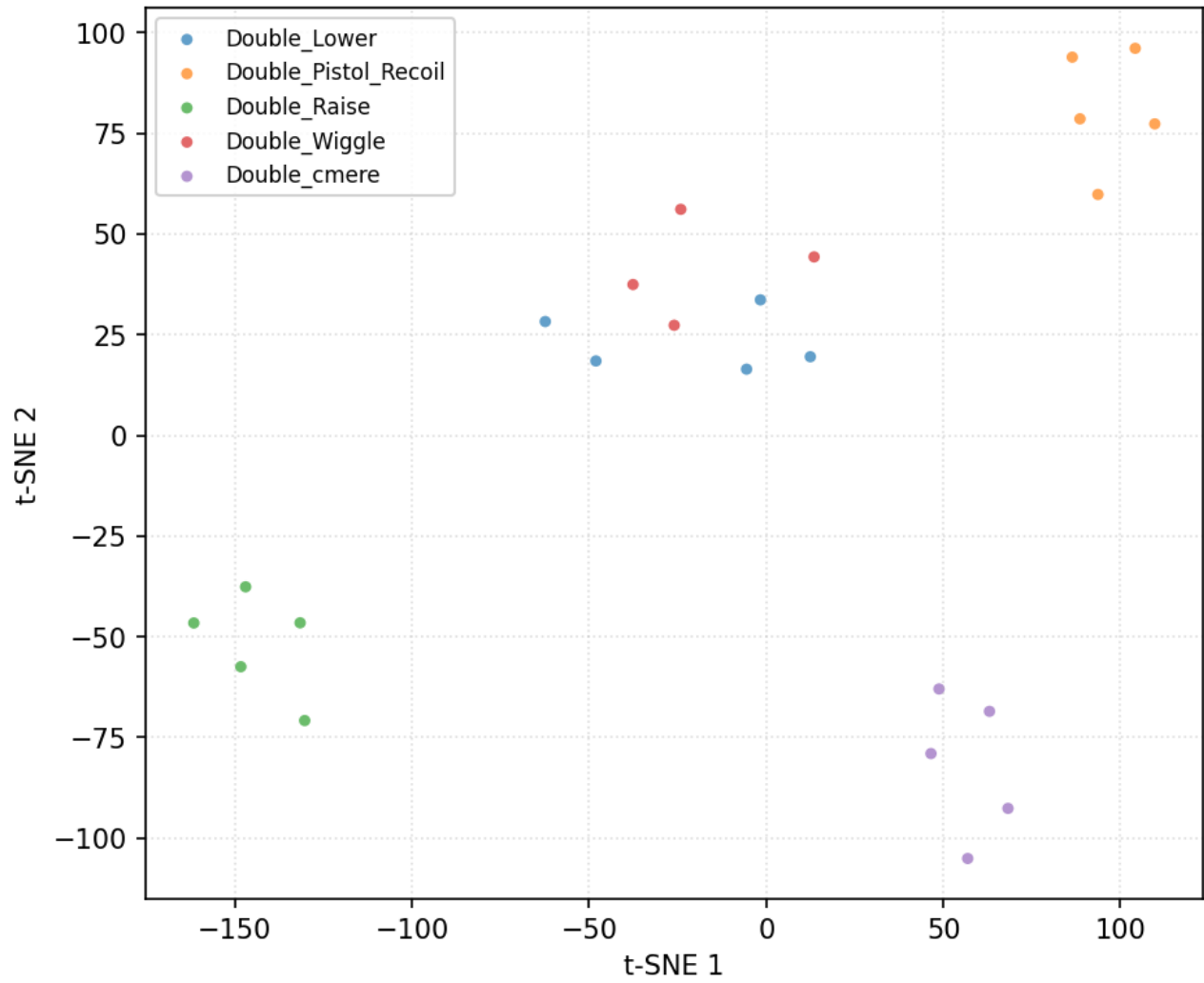


# Data diagnostics — user 8\_Jestin

24 trials, 5 classes (filtered + resampled, before per-trial normalisation).



t-SNE feature space projection



# Variant: BN\_GAP\_Wide

Conv1D(32,k=8)+BN→Conv1D(64,k=8)+BN→Conv1D(128,k=5)+BN→GAP→Dropout(0.3)→Softmax [LOUO leader]

|                  |                 |
|------------------|-----------------|
| Trainable params | 104,294         |
| LOUO mean ± std  | 0.9712 ± 0.0711 |
| Folds run        | 11              |
| Inner-val mode   | user            |
| Per-trial norm   | zscore          |

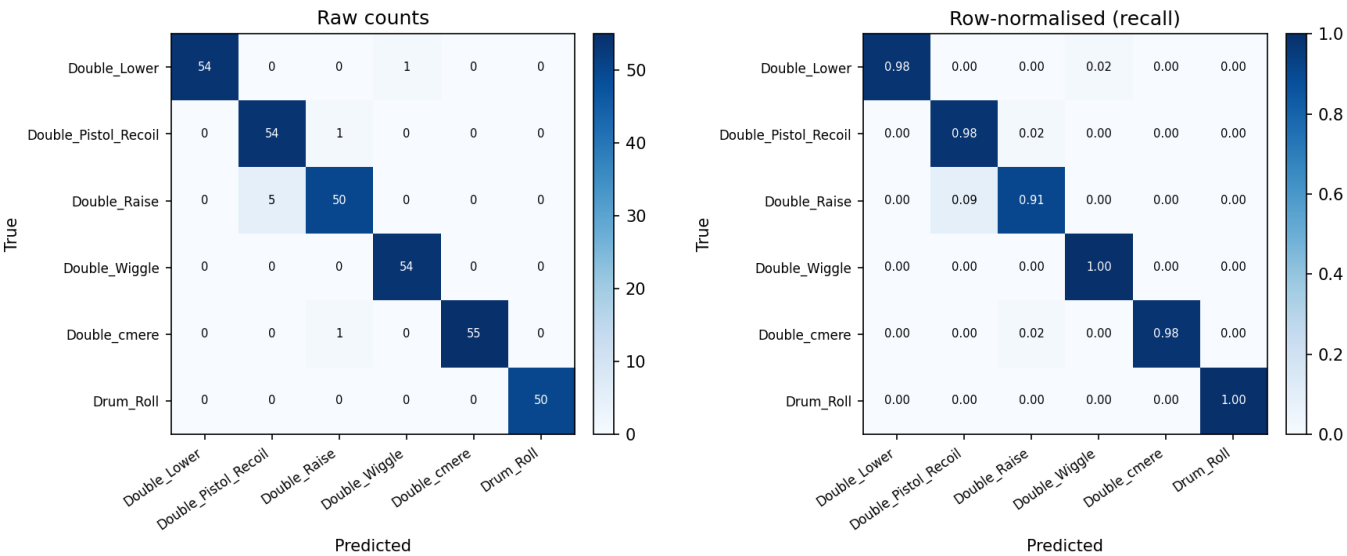
## LOUO per-fold results

| Fold | Held-out user | Inner-val user | n_train | n_test | Accuracy |
|------|---------------|----------------|---------|--------|----------|
| 1    | 11_Mansh      | 12_Marcus      | 795     | 30     | 1.0000   |
| 2    | 12_Marcus     | 14_StephenV2   | 795     | 30     | 1.0000   |
| 3    | 14_StephenV2  | 17_Raquel      | 795     | 30     | 1.0000   |
| 4    | 17_Raquel     | 18_Bridgette   | 795     | 30     | 1.0000   |
| 5    | 18_Bridgette  | 1_Alan         | 795     | 30     | 0.9667   |
| 6    | 1_Alan        | 2_Alex         | 795     | 30     | 0.9667   |
| 7    | 2_Alex        | 3_Anghad       | 792     | 30     | 1.0000   |
| 8    | 3_Anghad      | 4_Daniel       | 792     | 31     | 1.0000   |
| 9    | 4_Daniel      | 6_Harry        | 795     | 30     | 1.0000   |
| 10   | 6_Harry       | 8_Jestin       | 813     | 30     | 1.0000   |
| 11   | 8_Jestin      | 11_Mansh       | 813     | 24     | 0.7500   |

## Per-class metrics (aggregated across folds)

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 1.000     | 0.982  | 0.991 | 55      |
| Double_Pistol_Recoil | 0.915     | 0.982  | 0.947 | 55      |
| Double_Raise         | 0.962     | 0.909  | 0.935 | 55      |
| Double_Wiggle        | 0.982     | 1.000  | 0.991 | 54      |
| Double_cmere         | 1.000     | 0.982  | 0.991 | 56      |
| Drum_Roll            | 1.000     | 1.000  | 1.000 | 50      |
| macro avg            | 0.976     | 0.976  | 0.976 | 325     |
| weighted avg         | 0.976     | 0.975  | 0.975 | 325     |

## Confusion matrix (aggregated across folds)



Final retrain (deployment model)

|                      |                                       |
|----------------------|---------------------------------------|
| Variant              | BN_GAP_Wide                           |
| Held-out user        | — (trained on all users)              |
| Held-out accuracy    | —                                     |
| Training-set size    | 975 samples (325 before augmentation) |
| Trainable params     | 104,294                               |
| Final train loss     | 0.0002                                |
| Final train accuracy | 1.0000                                |
| Model path           | — (not saved)                         |
| Labels path          | — (not saved)                         |
| Scaler path          | — (per-trial norm in use)             |